TEMPEST: SGW 2023 Data Viz

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April 2023

```
#### April 2023 ###
##################################################################################################
#read in some data
dat <- read.csv("2023 Processed Data/COMPASS_TEMPEST_SGW_202304_Processed.csv")
head(dat)
```

```
##   X   Machine       User Run_Date Sample_Year Sample_Month Sample_Day
## 1 1 Varian GC Max Wegner 20231121        2023        April          6
## 2 2 Varian GC Max Wegner 20231121        2023        April          6
## 3 3 Varian GC Max Wegner 20231121        2023        April          6
## 4 4 Varian GC Max Wegner 20231121        2023        April          6
## 5 5 Varian GC Max Wegner 20231121        2023        April          6
## 6 6 Varian GC Max Wegner 20231121        2023        April          6
##   Sample_Time Time_Zone G_W Sample_Plot Sample_Grid Sample_Depth Sample_ID
## 1        1157       EDT   G     Control          C3           10         1
## 2        1158       EDT   G     Control          C3           20         2
## 3        1159       EDT   G     Control          C3           35         3
## 4        1200       EDT   G     Control          C3           50         4
## 5        1151       EDT   G     Control          C6           10         5
## 6        1152       EDT   G     Control          C6           20         6
##   Dilution_Factor Notes     CH4_Flag     CO2_Flag CH4_Conc_ppm_dilcorr
## 1               1    NA Within Range Within Range            1.3928376
## 2               1    NA Within Range Within Range            1.3539094
## 3               1    NA Within Range Within Range            0.6094086
## 4               1    NA Within Range Within Range            0.8251354
## 5               1    NA Within Range Within Range            1.6328945
## 6               1    NA Within Range Within Range            0.7862073
##   CO2_Conc_ppm_dilcorr CH4_Conc_umol CO2_Conc_umol
## 1             7035.928      86.83526      438648.9
## 2             8605.119      84.40832      536478.7
## 3            10791.868      37.99305      672809.7
## 4            10072.795      51.44236      627979.7
## 5             2879.572     101.80140      179524.4
## 6            15606.923      49.01541      973000.2
```

```
dat <-  dat %>%
  filter(!str_detect(Sample_Plot, "Transition"))
head(dat)
```

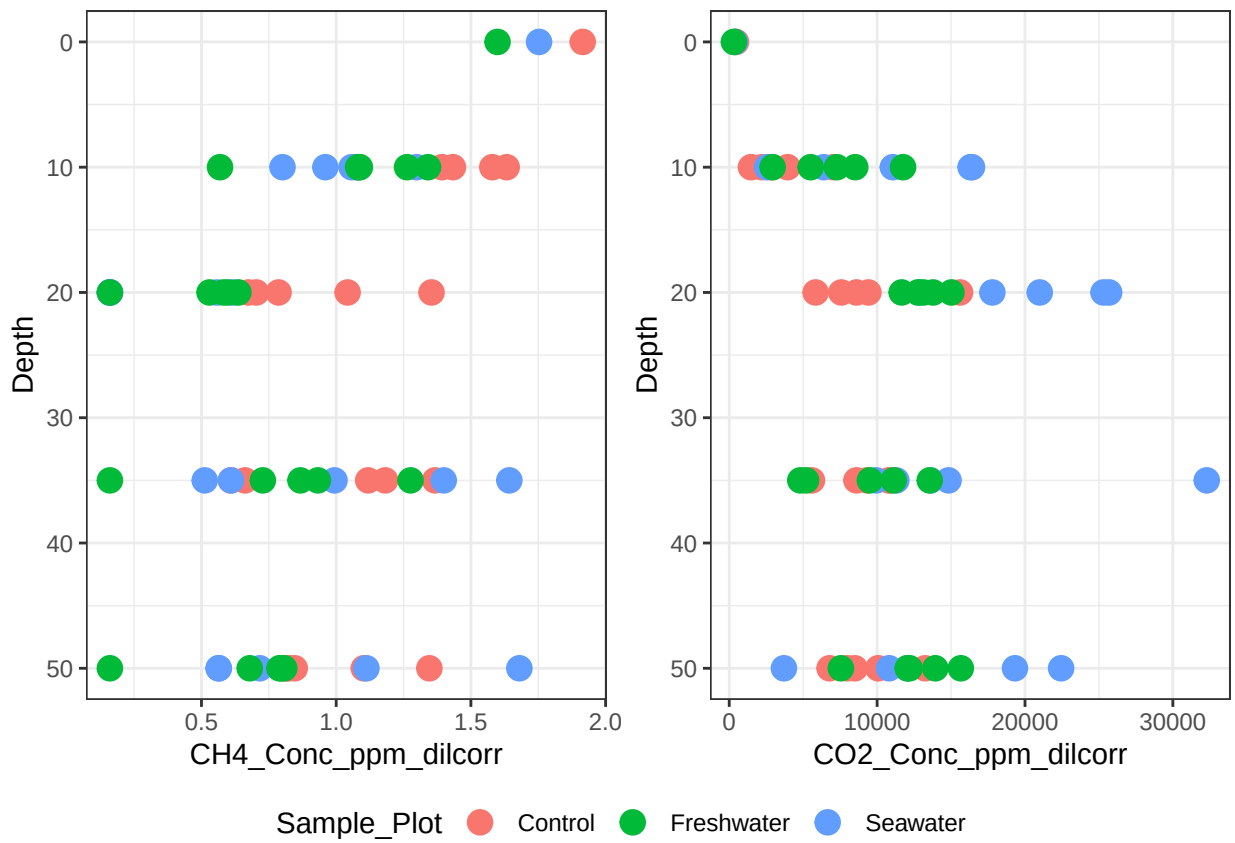
```
##   X   Machine       User Run_Date Sample_Year Sample_Month Sample_Day
## 1 1 Varian GC Max Wegner 20231121        2023        April          6
## 2 2 Varian GC Max Wegner 20231121        2023        April          6
## 3 3 Varian GC Max Wegner 20231121        2023        April          6
## 4 4 Varian GC Max Wegner 20231121        2023        April          6
## 5 5 Varian GC Max Wegner 20231121        2023        April          6
## 6 6 Varian GC Max Wegner 20231121        2023        April          6
##   Sample_Time Time_Zone G_W Sample_Plot Sample_Grid Sample_Depth Sample_ID
## 1        1157       EDT   G     Control          C3           10         1
## 2        1158       EDT   G     Control          C3           20         2
## 3        1159       EDT   G     Control          C3           35         3
## 4        1200       EDT   G     Control          C3           50         4
## 5        1151       EDT   G     Control          C6           10         5
## 6        1152       EDT   G     Control          C6           20         6
##   Dilution_Factor Notes     CH4_Flag     CO2_Flag CH4_Conc_ppm_dilcorr
## 1               1    NA Within Range Within Range            1.3928376
## 2               1    NA Within Range Within Range            1.3539094
## 3               1    NA Within Range Within Range            0.6094086
## 4               1    NA Within Range Within Range            0.8251354
## 5               1    NA Within Range Within Range            1.6328945
## 6               1    NA Within Range Within Range            0.7862073
##   CO2_Conc_ppm_dilcorr CH4_Conc_umol CO2_Conc_umol
## 1             7035.928      86.83526      438648.9
## 2             8605.119      84.40832      536478.7
## 3            10791.868      37.99305      672809.7
## 4            10072.795      51.44236      627979.7
## 5             2879.572     101.80140      179524.4
## 6            15606.923      49.01541      973000.2
```

```r
dat$Depth <- dat$Sample_Depth
dat$Depth <- revalue(dat$Depth, c("10" = 10, "20" = 20, "35" = 35, "50"= 50, "ATM" = 0))
dat$Depth <- as.numeric(dat$Depth)

soil_ch4 <- ggplot(data=dat, aes( x=CH4_Conc_ppm_dilcorr, y=Depth, color=Sample_Plot)) +
  geom_point(size=4)  + scale_y_reverse() + theme_bw()

soil_co2 <- ggplot(data=dat, aes( x=CO2_Conc_ppm_dilcorr, y=Depth, color=Sample_Plot)) +
  geom_point(size=4)  + scale_y_reverse() + theme_bw()

ggarrange(soil_ch4, soil_co2, nrow=1, ncol=2, common.legend = TRUE, legend="bottom")
```




```
dat$Sample_Month <- as.factor(dat$Sample_Month)
dat$Sample_Depth <- as.factor(dat$Sample_Depth)
dat$Sample_Plot <- as.factor(dat$Sample_Plot)

ghg_avg <- dat %>%
  group_by(Sample_Depth, Sample_Plot) %>%
  dplyr::summarize(mean_CH4_ppm = mean(CH4_Conc_ppm_dilcorr, na.rm=TRUE),
                   sd_CH4 = sd(CH4_Conc_ppm_dilcorr, na.rm=TRUE),
                   mean_CO2_ppm = mean(CO2_Conc_ppm_dilcorr, na.rm=TRUE),
                   sd_CO2 = sd(CO2_Conc_ppm_dilcorr, na.rm=TRUE))
```

```
## `summarise()` has grouped output by 'Sample_Depth'. You can override using the
## `.groups` argument.
```

```
head(ghg_avg)
```

```
## # A tibble: 6 x 6
## # Groups:   Sample_Depth [2]
##   Sample_Depth Sample_Plot mean_CH4_ppm sd_CH4 mean_CO2_ppm sd_CO2
##   <fct>        <fct>              <dbl>  <dbl>        <dbl>  <dbl>
## 1 10           Control            1.42   0.214        3507.  2174.
## 2 10           Freshwater         1.07   0.301        7199.  3300.
## 3 10           Seawater           1.04   0.182       10550.  6105.
## 4 20           Control            0.912  0.286        9409.  3712.
```

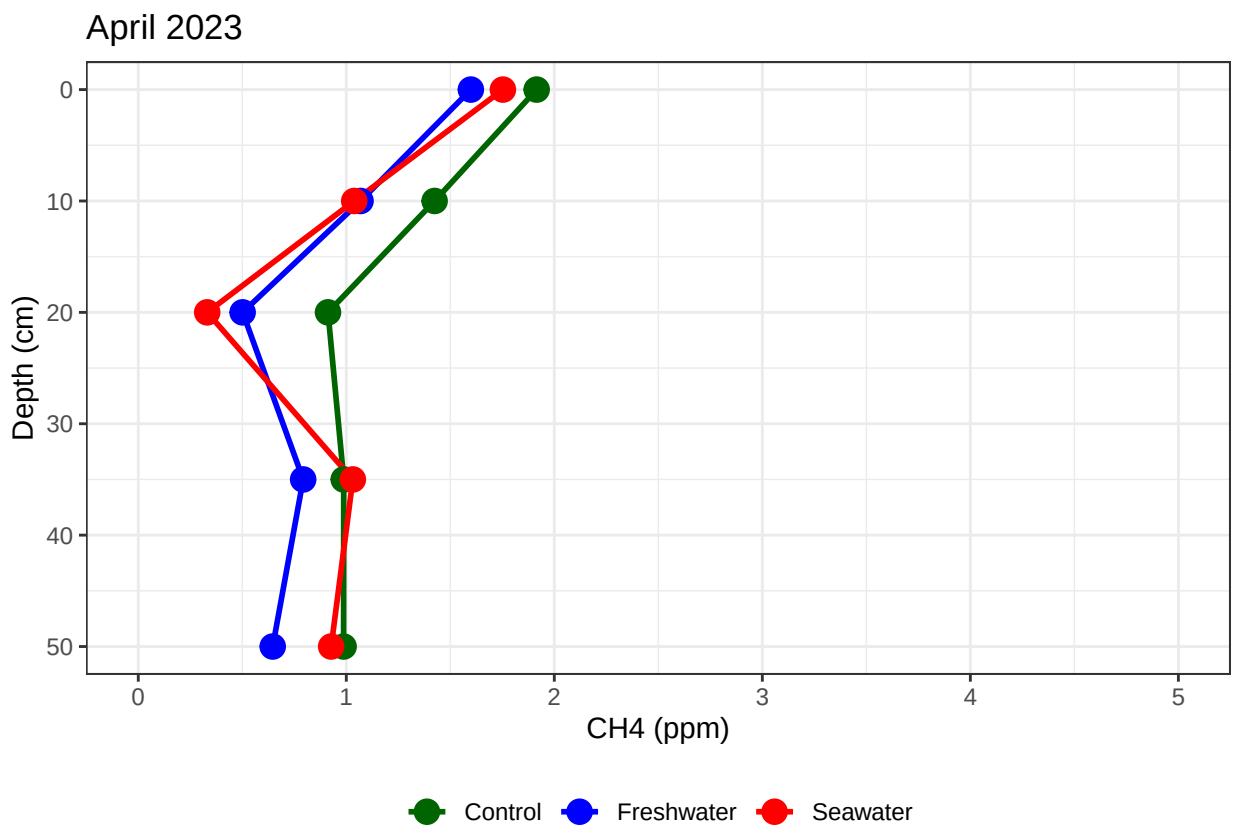
```
## 5 20           Freshwater         0.502  0.195       13275.  1246.
## 6 20           Seawater           0.331  0.236       20529.  5388.
```

```
ghg_avg$Depth <- revalue(ghg_avg$Sample_Depth, c("10" = 10, "20" = 20, "35" = 35, "50"= 50, "ATM" = 0))
ghg_avg$Depth <- as.character(ghg_avg$Depth)
ghg_avg$Depth <- as.numeric(ghg_avg$Depth)

april_avg_soil_ch4 <- ggplot(data=ghg_avg, aes(col=Sample_Plot, fill=Sample_Plot)) +
  geom_line(size=1, aes(Depth, mean_CH4_ppm)) +
  geom_point(size=4, aes(Depth, mean_CH4_ppm)) +
  scale_color_manual(values=c( "darkgreen","blue", "red")) +
  scale_fill_manual(values=c( "darkgreen","blue", "red")) +
  coord_flip() +
  scale_x_reverse() +
  ylim(0,5) +
  labs(title = "April 2023", y="CH4 (ppm)", x="Depth (cm)") +
  theme_bw() +
  theme(legend.position="bottom") +
  theme(legend.title=element_blank())
```

```
## Warning: Using ‘size‘ aesthetic for lines was deprecated in ggplot2 3.4.0.
## i Please use ‘linewidth‘ instead.
## This warning is displayed once every 8 hours.
## Call ‘lifecycle::last_lifecycle_warnings()‘ to see where this warning was
## generated.
```

```
april_avg_soil_ch4
```

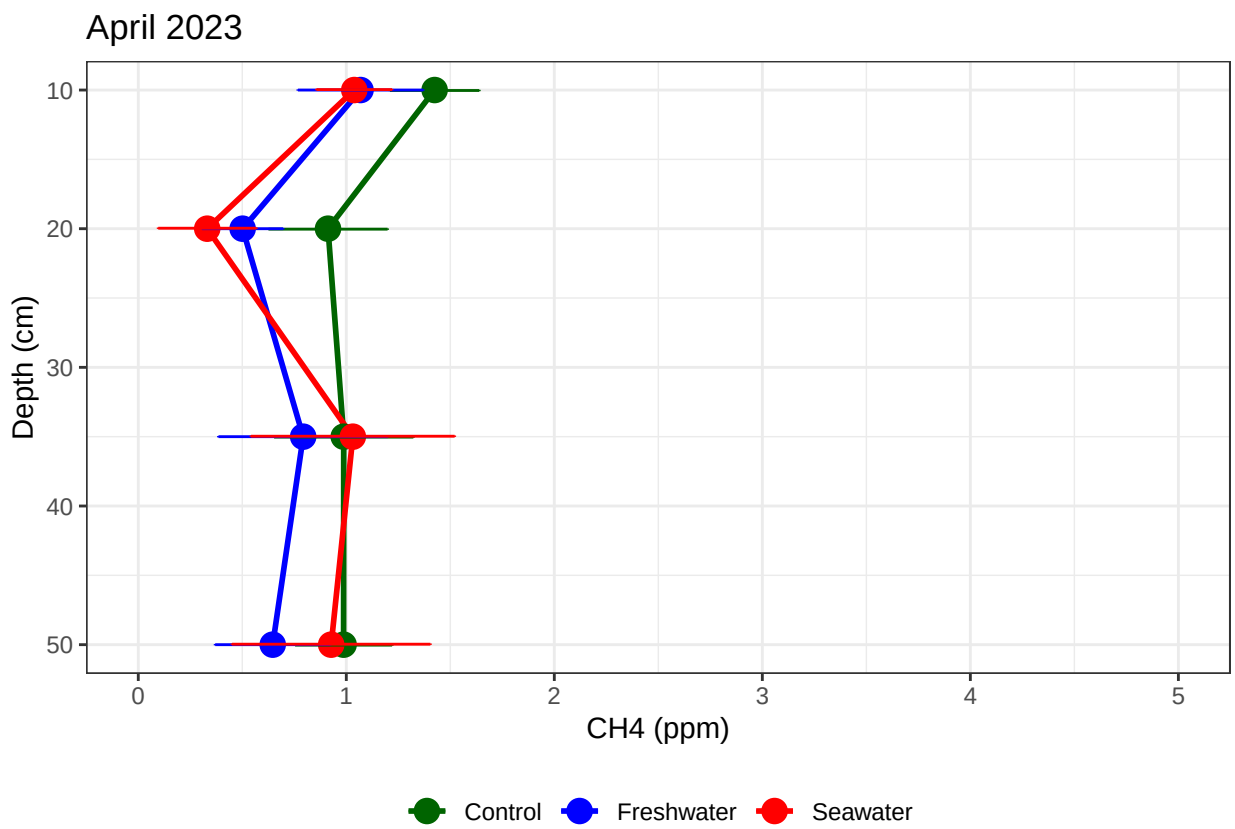



```
ghg_avg1 <-  ghg_avg %>%
  filter(!(Sample_Depth == "ATM"))
head(ghg_avg1)
```

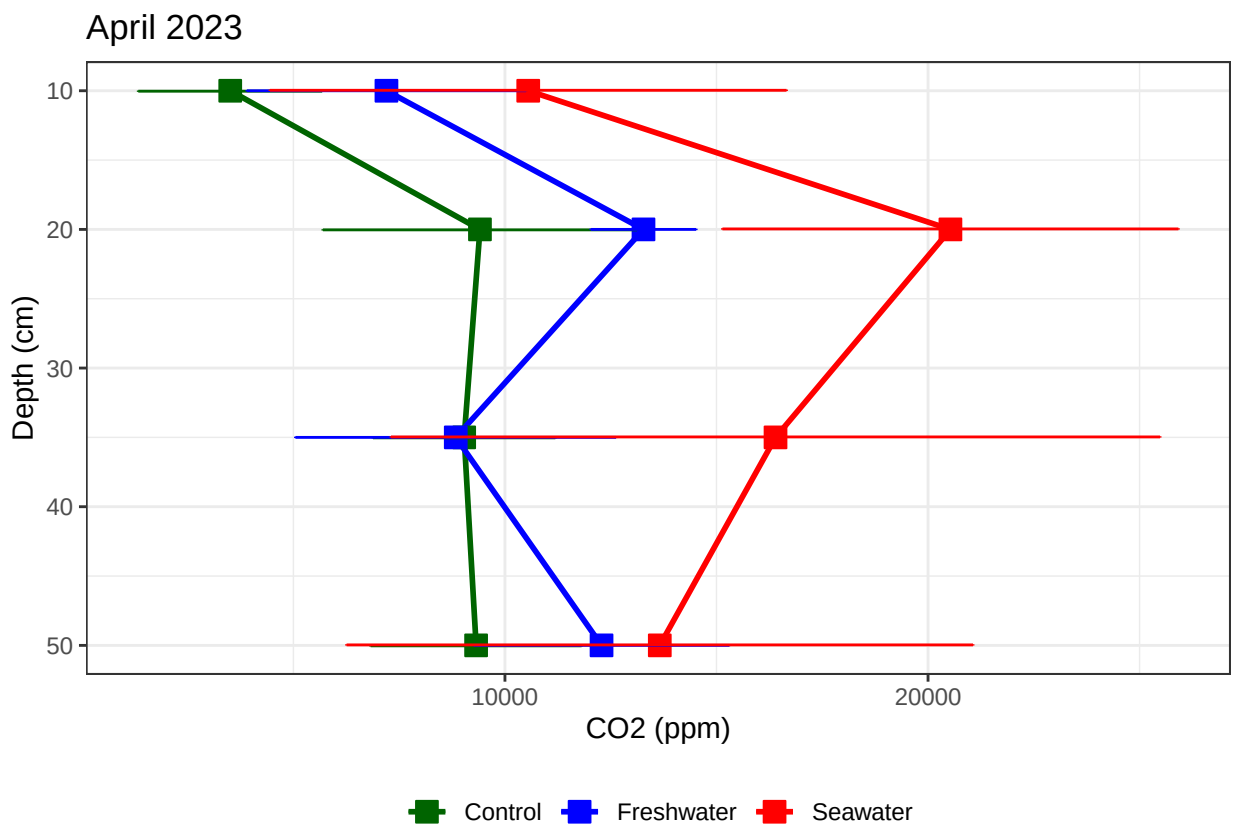
```
## # A tibble: 6 x 7
## # Groups:   Sample_Depth [2]
##   Sample_Depth Sample_Plot mean_CH4_ppm sd_CH4 mean_CO2_ppm sd_CO2 Depth
##   <fct>        <fct>              <dbl>  <dbl>        <dbl>  <dbl> <dbl>
## 1 10           Control            1.42   0.214        3507.  2174.    10
## 2 10           Freshwater         1.07   0.301        7199.  3300.    10
## 3 10           Seawater           1.04   0.182       10550.  6105.    10
## 4 20           Control            0.912  0.286        9409.  3712.    20
## 5 20           Freshwater         0.502  0.195       13275.  1246.    20
## 6 20           Seawater           0.331  0.236       20529.  5388.    20
```

```
april_avg_soil_ch41 <- ggplot(data=ghg_avg1, aes(col=Sample_Plot, fill=Sample_Plot)) +
  geom_line(size=1, aes(Depth, mean_CH4_ppm)) +
  geom_point(size=4, aes(Depth, mean_CH4_ppm)) +
  scale_color_manual(values=c( "darkgreen","blue", "red")) +
  scale_fill_manual(values=c( "darkgreen","blue", "red")) +
  coord_flip() +
  scale_x_reverse() +
  ylim(0,5) +
  labs(title = "April 2023", y="CH4 (ppm)", x="Depth (cm)") +
  theme_bw() +
```

```
  theme(legend.position="bottom") +
  theme(legend.title=element_blank()) +
  geom_errorbar(aes(Depth, ymin=mean_CH4_ppm-sd_CH4,
                    ymax=mean_CH4_ppm+sd_CH4),width=0.3,position=position_dodge(.1))
april_avg_soil_ch41
```




```
april_avg_soil_co21 <- ggplot(data=ghg_avg1, aes(col=Sample_Plot, fill=Sample_Plot)) +
  geom_line(size=1, aes(Depth, mean_CO2_ppm)) +
  geom_point(shape=22, size=4, aes(Depth, mean_CO2_ppm)) +
  scale_color_manual(values=c( "darkgreen","blue", "red")) +
  scale_fill_manual(values=c( "darkgreen","blue", "red")) +
  coord_flip() +
  scale_x_reverse() +
  #ylim(0,5) +
  labs(title = "April 2023", y="CO2 (ppm)", x="Depth (cm)") +
  theme_bw() +
  theme(legend.position="bottom") +
  theme(legend.title=element_blank())+
  geom_errorbar(aes(Depth, ymin=mean_CO2_ppm-sd_CO2,
                    ymax=mean_CO2_ppm+sd_CO2),width=0.3,position=position_dodge(.1))
april_avg_soil_co21
```




```
##########################################################################################
```

May 2023

```
#### May 2023 ####
##########################################################################################
#read in some data
dat <- read.csv("2023 Processed Data/COMPASS_TEMPEST_SGW_202305_Processed.csv")
head(dat)
```

```
##   X   Machine       User Run_Date Sample_Year Sample_Month Sample_Day
## 1 1 Varian GC Max Wegner 20231130        2023          May         16
## 2 2 Varian GC Max Wegner 20231130        2023          May         16
## 3 3 Varian GC Max Wegner 20231130        2023          May         16
## 4 4 Varian GC Max Wegner 20231130        2023          May         16
## 5 5 Varian GC Max Wegner 20231130        2023          May         16
## 6 6 Varian GC Max Wegner 20231130        2023          May         16
##   Sample_Time Time_Zone G_W Sample_Plot Sample_Grid Sample_Depth Sample_ID
## 1         929       EDT   G     Control          C3           10         1
## 2         931       EDT   G     Control          C3           20         2
## 3         933       EDT   G     Control          C3           35         3
## 4         935       EDT   G     Control          C3           50         4
## 5         923       EDT   G     Control          C6           10         5
```

```
## 6         925       EDT   G     Control          C6           20         6
##   Dilution_Factor                                          Field_Notes
## 1               1
## 2               1
## 3               1
## 4               1 a bit of condensation in syringe; used 0.45um filter
## 5               1
## 6               1
##   CO2_Conc_ppm     CH4_Flag     CO2_Flag CH4_Conc_ppm_dilcorr
## 1     5350.498 Within Range Within Range            1.8154719
## 2     8133.159 Within Range Within Range            1.1965048
## 3    12067.357 Within Range Within Range            0.9359702
## 4    15252.297 Within Range Within Range            0.8759683
## 5     6102.130 Within Range Within Range            1.6307292
## 6    21576.577 Within Range Within Range            0.5917487
##   CO2_Conc_ppm_dilcorr CH4_Conc_umol CO2_Conc_umol
## 1             5350.498     113.18403      333572.2
## 2             8133.159      74.59506      507054.8
## 3            12067.357      58.35226      752329.0
## 4            15252.297      54.61149      950891.3
## 5             6102.130     101.66641      380432.1
## 6            21576.577      36.89206     1345173.1
```

```r
dat <-  dat %>%
  filter(!str_detect(Sample_Plot, "Transition"))
head(dat)
```

```
##   X   Machine       User Run_Date Sample_Year Sample_Month Sample_Day
## 1 1 Varian GC Max Wegner 20231130        2023          May         16
## 2 2 Varian GC Max Wegner 20231130        2023          May         16
## 3 3 Varian GC Max Wegner 20231130        2023          May         16
## 4 4 Varian GC Max Wegner 20231130        2023          May         16
## 5 5 Varian GC Max Wegner 20231130        2023          May         16
## 6 6 Varian GC Max Wegner 20231130        2023          May         16
##   Sample_Time Time_Zone G_W Sample_Plot Sample_Grid Sample_Depth Sample_ID
## 1         929       EDT   G     Control          C3           10         1
## 2         931       EDT   G     Control          C3           20         2
## 3         933       EDT   G     Control          C3           35         3
## 4         935       EDT   G     Control          C3           50         4
## 5         923       EDT   G     Control          C6           10         5
## 6         925       EDT   G     Control          C6           20         6
##   Dilution_Factor                                          Field_Notes
## 1               1
## 2               1
## 3               1
## 4               1 a bit of condensation in syringe; used 0.45um filter
## 5               1
## 6               1
##   CO2_Conc_ppm     CH4_Flag     CO2_Flag CH4_Conc_ppm_dilcorr
## 1     5350.498 Within Range Within Range            1.8154719
## 2     8133.159 Within Range Within Range            1.1965048
## 3    12067.357 Within Range Within Range            0.9359702
## 4    15252.297 Within Range Within Range            0.8759683
## 5     6102.130 Within Range Within Range            1.6307292
```

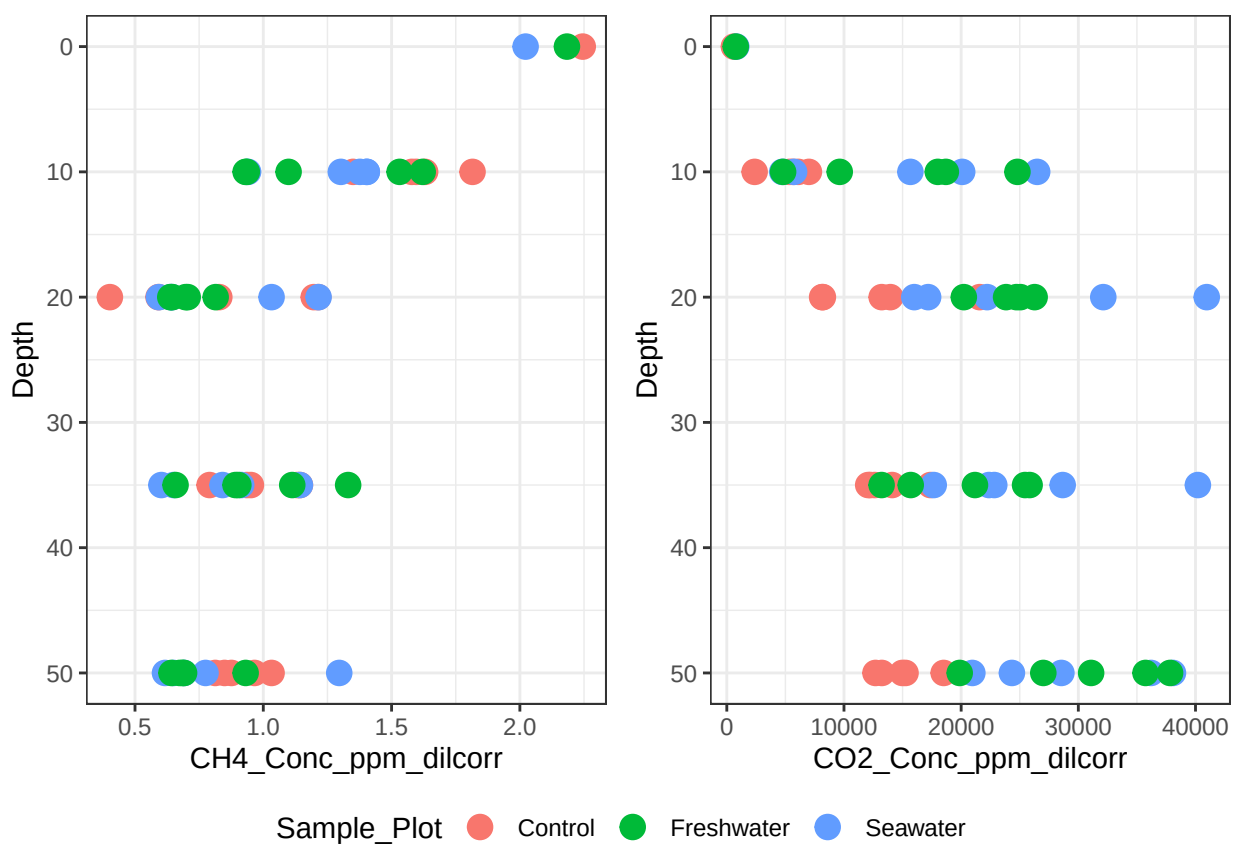
```
## 6    21576.577 Within Range Within Range            0.5917487
##   CO2_Conc_ppm_dilcorr CH4_Conc_umol CO2_Conc_umol
## 1             5350.498     113.18403      333572.2
## 2             8133.159      74.59506      507054.8
## 3            12067.357      58.35226      752329.0
## 4            15252.297      54.61149      950891.3
## 5             6102.130     101.66641      380432.1
## 6            21576.577      36.89206     1345173.1
```

```
dat$Depth <- dat$Sample_Depth
dat$Depth <- revalue(dat$Depth, c("10" = 10, "20" = 20, "35" = 35, "50"= 50, "ATM" = 0))
dat$Depth <- as.numeric(dat$Depth)

soil_ch4 <- ggplot(data=dat, aes( x=CH4_Conc_ppm_dilcorr, y=Depth, color=Sample_Plot)) +
  geom_point(size=4)  + scale_y_reverse() + theme_bw()

soil_co2 <- ggplot(data=dat, aes( x=CO2_Conc_ppm_dilcorr, y=Depth, color=Sample_Plot)) +
  geom_point(size=4)  + scale_y_reverse() + theme_bw()

ggarrange(soil_ch4, soil_co2, nrow=1, ncol=2, common.legend = TRUE, legend="bottom")
```




```
dat$Sample_Month <- as.factor(dat$Sample_Month)
dat$Sample_Depth <- as.factor(dat$Sample_Depth)
dat$Sample_Plot <- as.factor(dat$Sample_Plot)

levels(dat$Sample_Plot)
```

```
## [1] "Control"    "Freshwater" "Seawater"
```

```r
dat <- as.data.frame(dat)

ghg_avg <- dat %>%
  group_by(Sample_Depth, Sample_Plot) %>%
  dplyr::summarize(mean_CH4_ppm = mean(CH4_Conc_ppm_dilcorr, na.rm=TRUE),
              sd_CH4 = sd(CH4_Conc_ppm_dilcorr, na.rm=TRUE),
              mean_CO2_ppm = mean(CO2_Conc_ppm_dilcorr, na.rm=TRUE),
              sd_CO2 = sd(CO2_Conc_ppm_dilcorr, na.rm=TRUE))
```

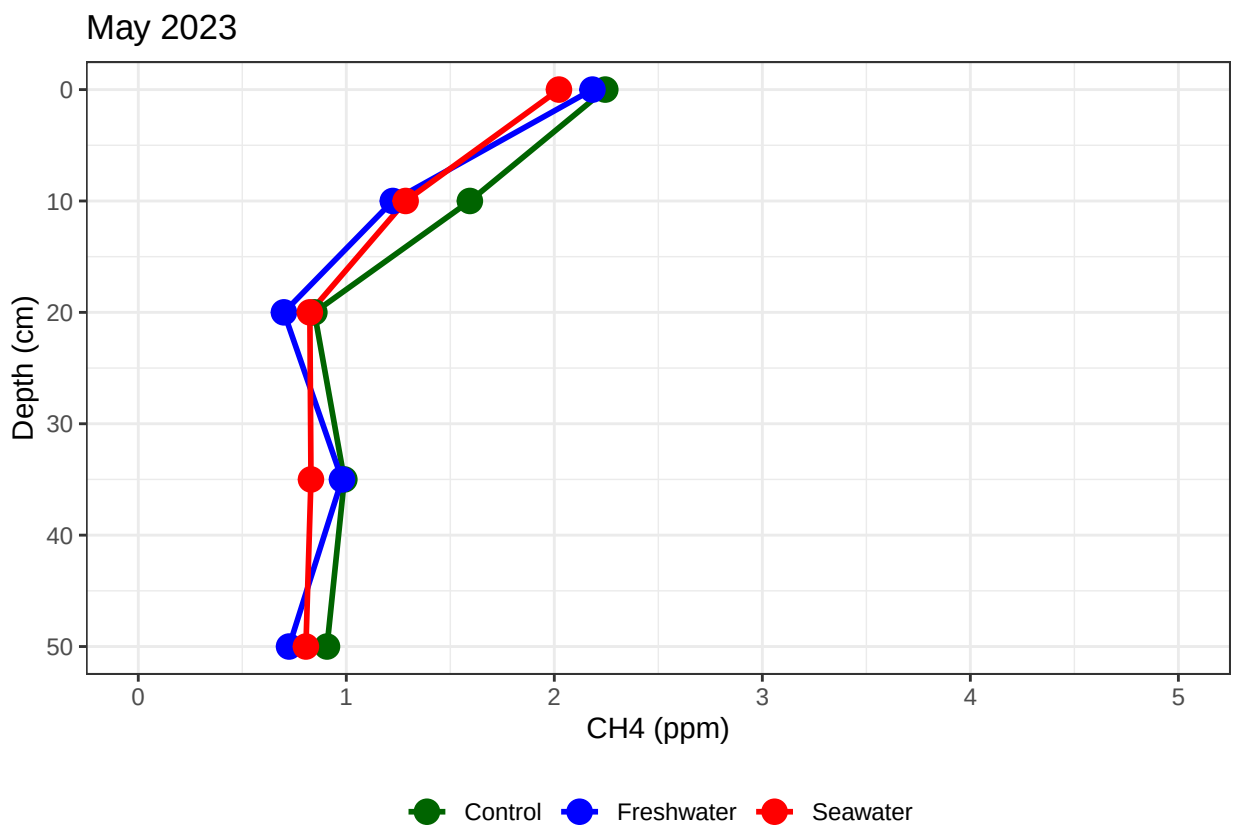
```
## `summarise()` has grouped output by 'Sample_Depth'. You can override using the
## `.groups` argument.
```

```r
head(ghg_avg)
```

```
## # A tibble: 6 x 6
## # Groups:   Sample_Depth [2]
##   Sample_Depth Sample_Plot mean_CH4_ppm sd_CH4 mean_CO2_ppm sd_CO2
##   <fct>        <fct>              <dbl>  <dbl>        <dbl>  <dbl>
## 1 10           Control            1.59  0.166         5297.  1749.
## 2 10           Freshwater         1.22  0.330        15188.  7927.
## 3 10           Seawater           1.29  0.197        14543.  9334.
## 4 20           Control            0.847 0.361        13017.  5506.
## 5 20           Freshwater         0.700 0.0707       24021.  2294.
## 6 20           Seawater           0.825 0.281        25697. 10641.
```

```r
ghg_avg$Depth <- revalue(ghg_avg$Sample_Depth, c("10" = 10, "20" = 20, "35" = 35, "50"= 50, "ATM" = 0))
ghg_avg$Depth <- as.character(ghg_avg$Depth)
ghg_avg$Depth <- as.numeric(ghg_avg$Depth)

may_avg_soil_ch4 <- ggplot(data=ghg_avg, aes(col=Sample_Plot, fill=Sample_Plot)) +
  geom_line(size=1, aes(Depth, mean_CH4_ppm)) +
  geom_point(size=4, aes(Depth, mean_CH4_ppm)) +
  scale_color_manual(values=c( "darkgreen","blue", "red")) +
  scale_fill_manual(values=c( "darkgreen","blue", "red")) +
  coord_flip() +
  scale_x_reverse() +
  ylim(0,5) +
  labs(title = "May 2023", y="CH4 (ppm)", x="Depth (cm)") +
  theme_bw() +
  theme(legend.position="bottom") +
  theme(legend.title=element_blank())
may_avg_soil_ch4
```

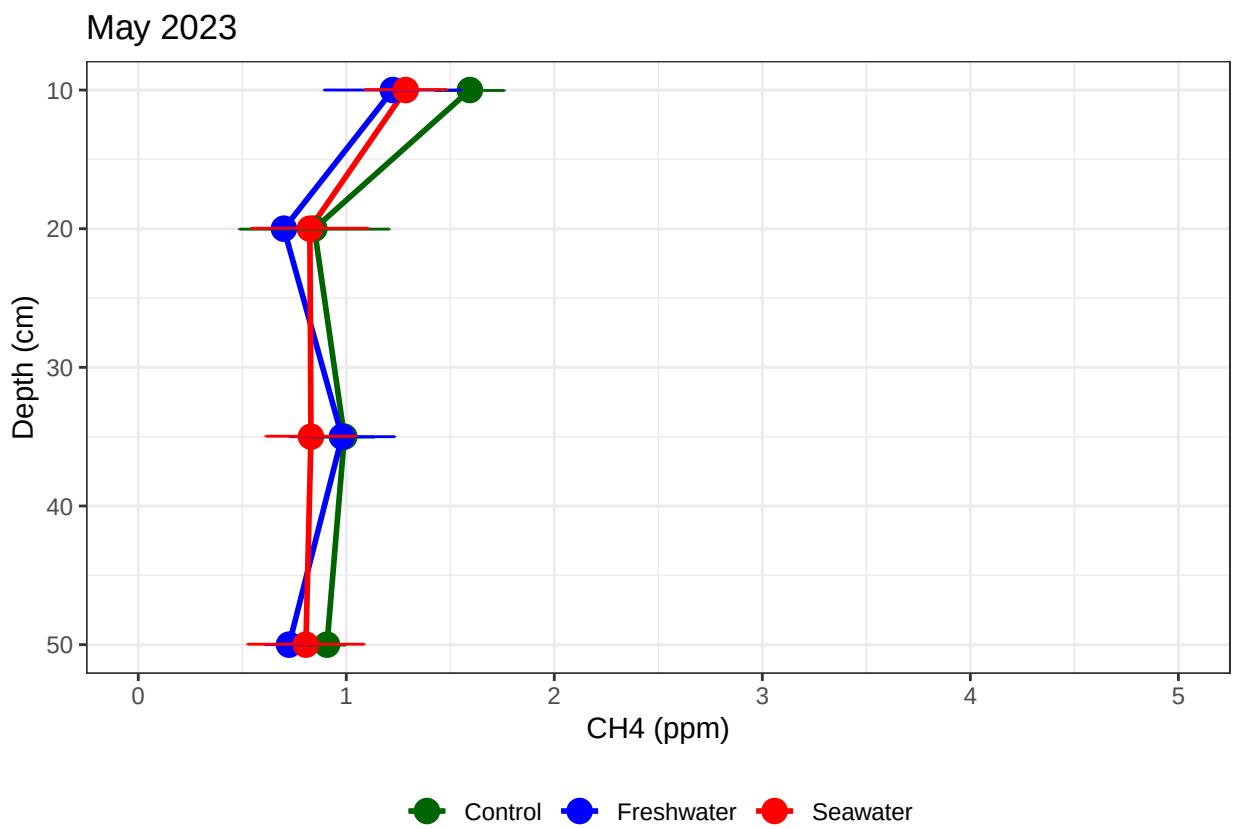



```
ghg_avg1 <-  ghg_avg %>%
  filter(!(Sample_Depth == "ATM"))
head(ghg_avg1)
```

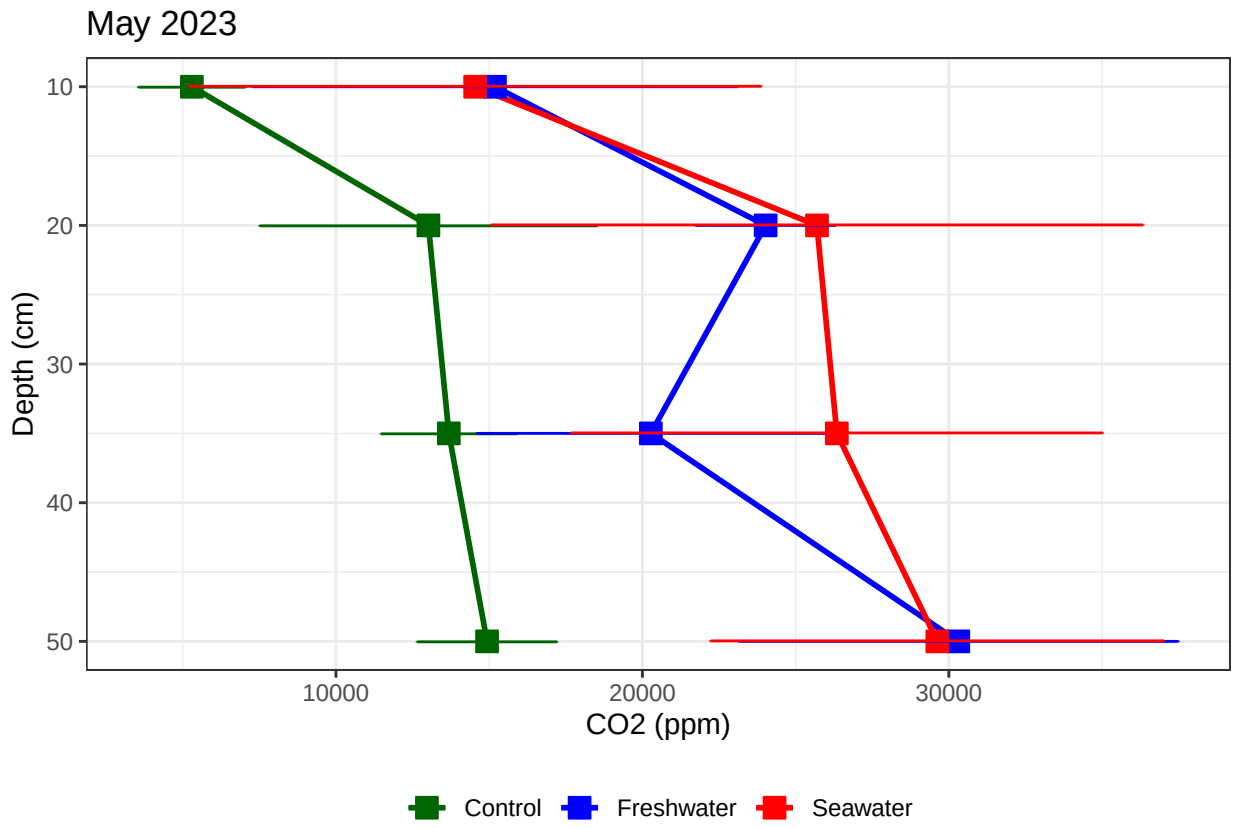
```
## # A tibble: 6 x 7
## # Groups:   Sample_Depth [2]
##   Sample_Depth Sample_Plot mean_CH4_ppm sd_CH4 mean_CO2_ppm sd_CO2 Depth
##   <fct>        <fct>              <dbl>  <dbl>        <dbl>  <dbl> <dbl>
## 1 10           Control            1.59  0.166         5297.  1749.    10
## 2 10           Freshwater         1.22  0.330        15188.  7927.    10
## 3 10           Seawater           1.29  0.197        14543.  9334.    10
## 4 20           Control            0.847 0.361        13017.  5506.    20
## 5 20           Freshwater         0.700 0.0707       24021.  2294.    20
## 6 20           Seawater           0.825 0.281        25697. 10641.    20
```

```
may_avg_soil_ch41 <- ggplot(data=ghg_avg1, aes(col=Sample_Plot, fill=Sample_Plot)) +
  geom_line(size=1, aes(Depth, mean_CH4_ppm)) +
  geom_point(size=4, aes(Depth, mean_CH4_ppm)) +
  scale_color_manual(values=c( "darkgreen","blue", "red")) +
  scale_fill_manual(values=c( "darkgreen","blue", "red")) +
  coord_flip() +
  scale_x_reverse() +
  ylim(0,5) +
  labs(title = "May 2023", y="CH4 (ppm)", x="Depth (cm)") +
  theme_bw() +
```

```
  theme(legend.position="bottom") +
  theme(legend.title=element_blank())+
  geom_errorbar(aes(Depth, ymin=mean_CH4_ppm-sd_CH4,
                    ymax=mean_CH4_ppm+sd_CH4),width=0.3,position=position_dodge(.1))
may_avg_soil_ch41
```




```
may_avg_soil_co21 <- ggplot(data=ghg_avg1, aes(col=Sample_Plot, fill=Sample_Plot)) +
  geom_line(size=1, aes(Depth, mean_CO2_ppm)) +
  geom_point(shape=22, size=4, aes(Depth, mean_CO2_ppm)) +
  scale_color_manual(values=c( "darkgreen","blue", "red")) +
  scale_fill_manual(values=c( "darkgreen","blue", "red")) +
  coord_flip() +
  scale_x_reverse() +
  #ylim(0,5) +
  labs(title = "May 2023", y="CO2 (ppm)", x="Depth (cm)") +
  theme_bw() +
  theme(legend.position="bottom") +
  theme(legend.title=element_blank())+
  geom_errorbar(aes(Depth, ymin=mean_CO2_ppm-sd_CO2,
                    ymax=mean_CO2_ppm+sd_CO2),width=0.3,position=position_dodge(.1))
may_avg_soil_co21
```




```
##################################################################################################
```

June 2023

```
#### June 2023 ###
##################################################################################################
#read in some data
dat <- read.csv("2023 Processed Data/COMPASS_TEMPEST_SGW_202306_Processed.csv")
head(dat)
```

```
##   X   Machine       User Run_Date Sample_Year Sample_Month Sample_Day
## 1 1 Varian GC Max Wegner 20230627        2023         June          4
## 2 2 Varian GC Max Wegner 20230627        2023         June          4
## 3 3 Varian GC Max Wegner 20230627        2023         June          4
## 4 4 Varian GC Max Wegner 20230627        2023         June          4
## 5 5 Varian GC Max Wegner 20230627        2023         June          4
## 6 6 Varian GC Max Wegner 20230627        2023         June          4
##   Event_Stamp Sample_Time Time_Zone G_W Sample_Plot Sample_Grid Sample_Depth
## 1           0        1445       EDT   G     Control          C3           10
## 2           0        1446       EDT   G     Control          C3           20
## 3           0        1448       EDT   G     Control          C3           35
## 4           0        1449       EDT   G     Control          C3           50
## 5           0        1451       EDT   G     Control          C6           10
```

```
## 6           0        1453       EDT   G     Control          C6           20
##   Sample_ID Dilution_Factor Field.Notes CH4_Conc_ppm CO2_Conc_ppm     CH4_Flag
## 1         1               1                1.4909905     5350.648 Within Range
## 2         2               1                1.1191802     8460.576 Within Range
## 3         3               1                0.8791507    11944.479 Within Range
## 4         4               1                0.7442322    19579.334 Within Range
## 5         5               1                1.6463036     5296.533 Within Range
## 6         6               1                1.6149272    20221.389 Within Range
##       CO2_Flag CH4_Conc_ppm_dilcorr CO2_Conc_ppm_dilcorr CH4_Conc_umol
## 1 Within Range            1.4909905             5350.648  0.0012188171
## 2 Within Range            1.1191802             8460.576  0.0009148790
## 3 Within Range            0.8791507            11944.479  0.0007186658
## 4 Within Range            0.7442322            19579.334  0.0006083761
## 5 Within Range            1.6463036             5296.533  0.0013457786
## 6 Within Range            1.6149272            20221.389  0.0013201298
##   CO2_Conc_umol CH4_Conc_nmol CO2_Conc_nmol
## 1      4.373912     1.2188171      4373.912
## 2      6.916137     0.9148790      6916.137
## 3      9.764070     0.7186658      9764.070
## 4     16.005218     0.6083761     16005.218
## 5      4.329676     1.3457786      4329.676
## 6     16.530068     1.3201298     16530.068
```

```
dat <-  dat %>%
  filter(!str_detect(Sample_Plot, "Transition"))
head(dat)
```

```
##   X   Machine       User Run_Date Sample_Year Sample_Month Sample_Day
## 1 1 Varian GC Max Wegner 20230627        2023         June          4
## 2 2 Varian GC Max Wegner 20230627        2023         June          4
## 3 3 Varian GC Max Wegner 20230627        2023         June          4
## 4 4 Varian GC Max Wegner 20230627        2023         June          4
## 5 5 Varian GC Max Wegner 20230627        2023         June          4
## 6 6 Varian GC Max Wegner 20230627        2023         June          4
##   Event_Stamp Sample_Time Time_Zone G_W Sample_Plot Sample_Grid Sample_Depth
## 1           0        1445       EDT   G     Control          C3           10
## 2           0        1446       EDT   G     Control          C3           20
## 3           0        1448       EDT   G     Control          C3           35
## 4           0        1449       EDT   G     Control          C3           50
## 5           0        1451       EDT   G     Control          C6           10
## 6           0        1453       EDT   G     Control          C6           20
##   Sample_ID Dilution_Factor Field.Notes CH4_Conc_ppm CO2_Conc_ppm     CH4_Flag
## 1         1               1                1.4909905     5350.648 Within Range
## 2         2               1                1.1191802     8460.576 Within Range
## 3         3               1                0.8791507    11944.479 Within Range
## 4         4               1                0.7442322    19579.334 Within Range
## 5         5               1                1.6463036     5296.533 Within Range
## 6         6               1                1.6149272    20221.389 Within Range
##       CO2_Flag CH4_Conc_ppm_dilcorr CO2_Conc_ppm_dilcorr CH4_Conc_umol
## 1 Within Range            1.4909905             5350.648  0.0012188171
## 2 Within Range            1.1191802             8460.576  0.0009148790
## 3 Within Range            0.8791507            11944.479  0.0007186658
## 4 Within Range            0.7442322            19579.334  0.0006083761
## 5 Within Range            1.6463036             5296.533  0.0013457786
```

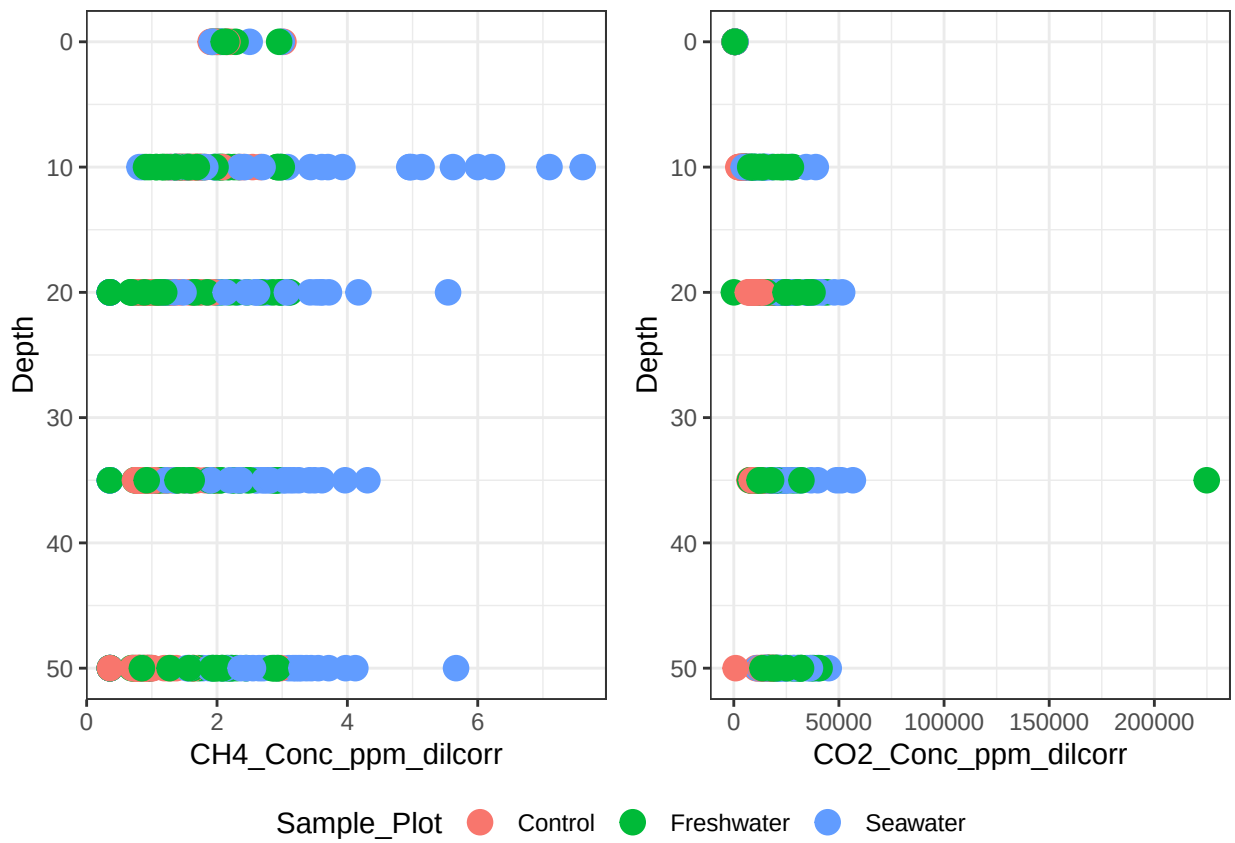
```
## 6 Within Range            1.6149272            20221.389  0.0013201298
##   CO2_Conc_umol CH4_Conc_nmol CO2_Conc_nmol
## 1      4.373912     1.2188171      4373.912
## 2      6.916137     0.9148790      6916.137
## 3      9.764070     0.7186658      9764.070
## 4     16.005218     0.6083761     16005.218
## 5      4.329676     1.3457786      4329.676
## 6     16.530068     1.3201298     16530.068
```

```
dat$Depth <- dat$Sample_Depth
dat$Depth <- revalue(dat$Depth, c("10" = 10, "20" = 20, "35" = 35, "50"= 50, "ATM" = 0))
dat$Depth <- as.numeric(dat$Depth)

soil_ch4 <- ggplot(data=dat, aes( x=CH4_Conc_ppm_dilcorr, y=Depth, color=Sample_Plot)) +
  geom_point(size=4)  + scale_y_reverse() + theme_bw()

soil_co2 <- ggplot(data=dat, aes( x=CO2_Conc_ppm_dilcorr, y=Depth, color=Sample_Plot)) +
  geom_point(size=4)  + scale_y_reverse() + theme_bw()

ggarrange(soil_ch4, soil_co2, nrow=1, ncol=2, common.legend = TRUE, legend="bottom")
```




```
dat$Sample_Month <- as.factor(dat$Sample_Month)
dat$Sample_Depth <- as.factor(dat$Sample_Depth)
dat$Sample_Plot <- as.factor(dat$Sample_Plot)
dat$Event_Stamp <- as.factor(dat$Event_Stamp)
```

```r
ghg_avg <- dat %>%
  group_by(Sample_Depth, Sample_Plot, Event_Stamp) %>%
  dplyr::summarize(mean_CH4_ppm = mean(CH4_Conc_ppm_dilcorr, na.rm=TRUE),
                   sd_CH4 = sd(CH4_Conc_ppm_dilcorr, na.rm=TRUE),
                   mean_CO2_ppm = mean(CO2_Conc_ppm_dilcorr, na.rm=TRUE),
                   sd_CO2 = sd(CO2_Conc_ppm_dilcorr, na.rm=TRUE))
```

```
## `summarise()` has grouped output by 'Sample_Depth', 'Sample_Plot'. You can
## override using the `.groups` argument.
```

```r
head(ghg_avg)
```

```
## # A tibble: 6 x 7
## # Groups:   Sample_Depth, Sample_Plot [1]
##   Sample_Depth Sample_Plot Event_Stamp mean_CH4_ppm sd_CH4 mean_CO2_ppm sd_CO2
##   <fct>        <fct>       <fct>              <dbl>  <dbl>        <dbl>  <dbl>
## 1 10           Control     0                   1.49  0.137        4767.  1357.
## 2 10           Control     1                   1.42  0.179        4134.   678.
## 3 10           Control     2                   1.60  0.102        4196.   865.
## 4 10           Control     3_AM                1.64  0.155        4554.  1136.
## 5 10           Control     3_PM                2.41  0.219        6187.  1151.
## 6 10           Control     4                   1.62  0.173        4329.  1004.
```

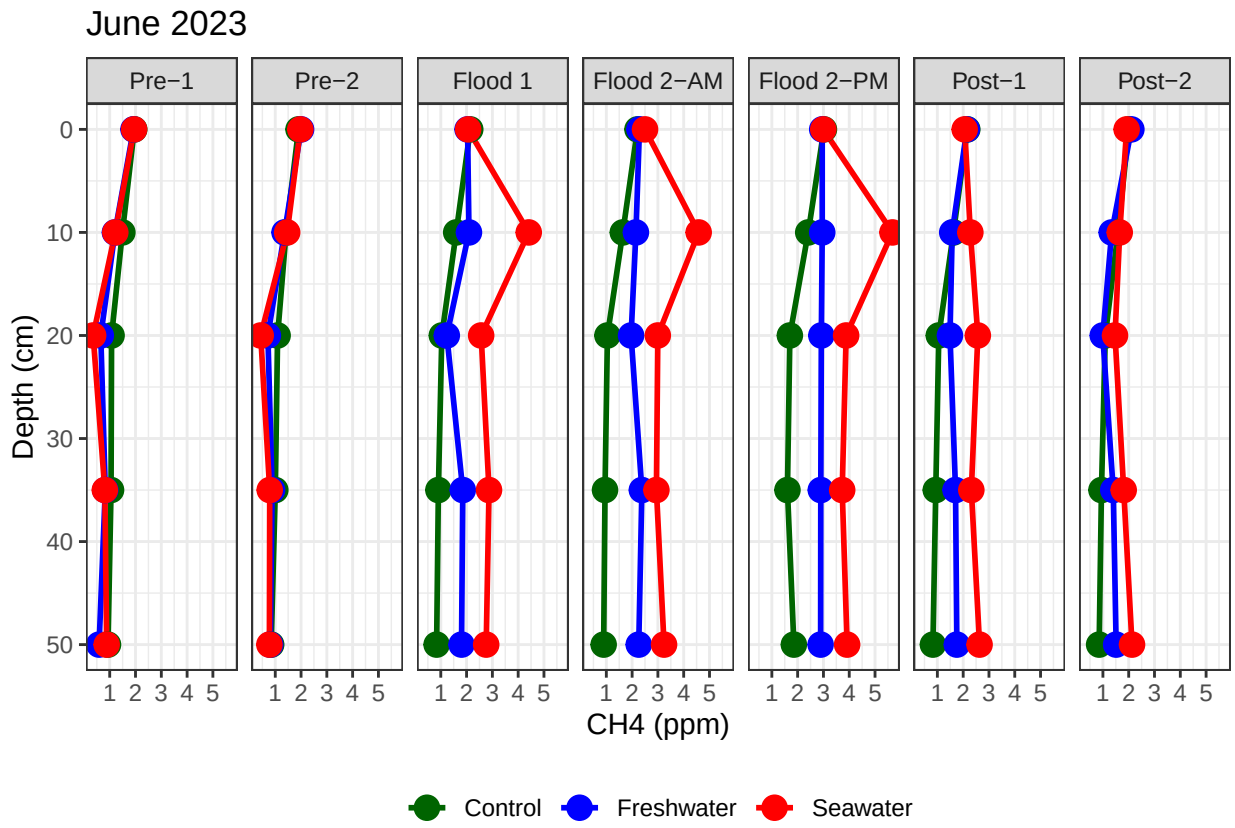
```r
ghg_avg$Depth <- revalue(ghg_avg$Sample_Depth, c("10" = 10, "20" = 20, "35" = 35, "50"= 50, "ATM" = 0))
ghg_avg$Depth <- as.character(ghg_avg$Depth)
ghg_avg$Depth <- as.numeric(ghg_avg$Depth)

levels(ghg_avg$Event_Stamp)
```

```
## [1] "0"    "1"    "2"    "3_AM" "3_PM" "4"    "5"
```

```r
ghg_avg$Event <- revalue(ghg_avg$Event_Stamp, c("0" = "Pre-1", "1" = "Pre-2", "2" = "Flood 1",
                                                "3_AM"= "Flood 2-AM", "3_PM" = "Flood 2-PM", "4" = "Po

june_avg_soil_ch4 <- ggplot(data=ghg_avg, aes(col=Sample_Plot, fill=Sample_Plot)) +
  geom_line(size=1, aes(Depth, mean_CH4_ppm)) +
  geom_point(size=4, aes(Depth, mean_CH4_ppm)) +
  scale_color_manual(values=c( "darkgreen","blue", "red")) +
  scale_fill_manual(values=c( "darkgreen","blue", "red")) +
  coord_flip() +
  scale_x_reverse() +
  facet_grid(~factor(Event, levels=c("Pre-1", "Pre-2", "Flood 1","Flood 2-AM" ,"Flood 2-PM", "Post-1","
  labs(title = "June 2023", y="CH4 (ppm)", x="Depth (cm)") +
  theme_bw() +
  theme(legend.position="bottom") +
  theme(legend.title=element_blank())
june_avg_soil_ch4
```

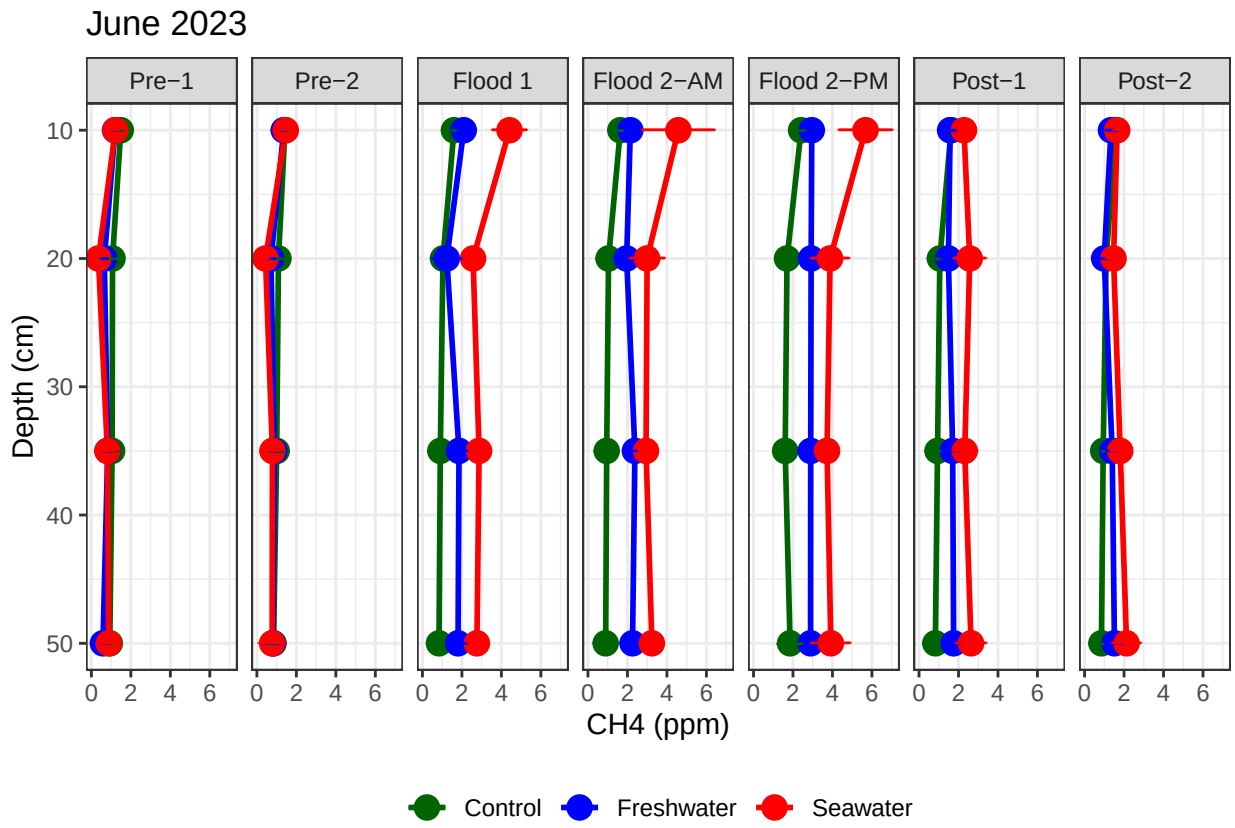



```
ghg_avg1 <-  ghg_avg %>%
  filter(!(Sample_Depth == "ATM"))
head(ghg_avg1)
```

```
## # A tibble: 6 x 9
## # Groups:   Sample_Depth, Sample_Plot [1]
##   Sample_Depth Sample_Plot Event_Stamp mean_CH4_ppm sd_CH4 mean_CO2_ppm sd_CO2
##   <fct>        <fct>       <fct>              <dbl>  <dbl>        <dbl>  <dbl>
## 1 10           Control     0                   1.49  0.137        4767.  1357.
## 2 10           Control     1                   1.42  0.179        4134.   678.
## 3 10           Control     2                   1.60  0.102        4196.   865.
## 4 10           Control     3_AM                1.64  0.155        4554.  1136.
## 5 10           Control     3_PM                2.41  0.219        6187.  1151.
## 6 10           Control     4                   1.62  0.173        4329.  1004.
## # i 2 more variables: Depth <dbl>, Event <fct>
```

```
june_avg_soil_ch41 <- ggplot(data=ghg_avg1, aes(col=Sample_Plot, fill=Sample_Plot)) +
  geom_line(size=1, aes(Depth, mean_CH4_ppm)) +
  geom_point(size=4, aes(Depth, mean_CH4_ppm)) +
  scale_color_manual(values=c( "darkgreen","blue", "red")) +
  scale_fill_manual(values=c( "darkgreen","blue", "red")) +
  coord_flip() +
  scale_x_reverse() +
  facet_grid(~factor(Event, levels=c("Pre-1", "Pre-2", "Flood 1","Flood 2-AM" ,"Flood 2-PM", "Post-1","
  labs(title = "June 2023", y="CH4 (ppm)", x="Depth (cm)") +
```

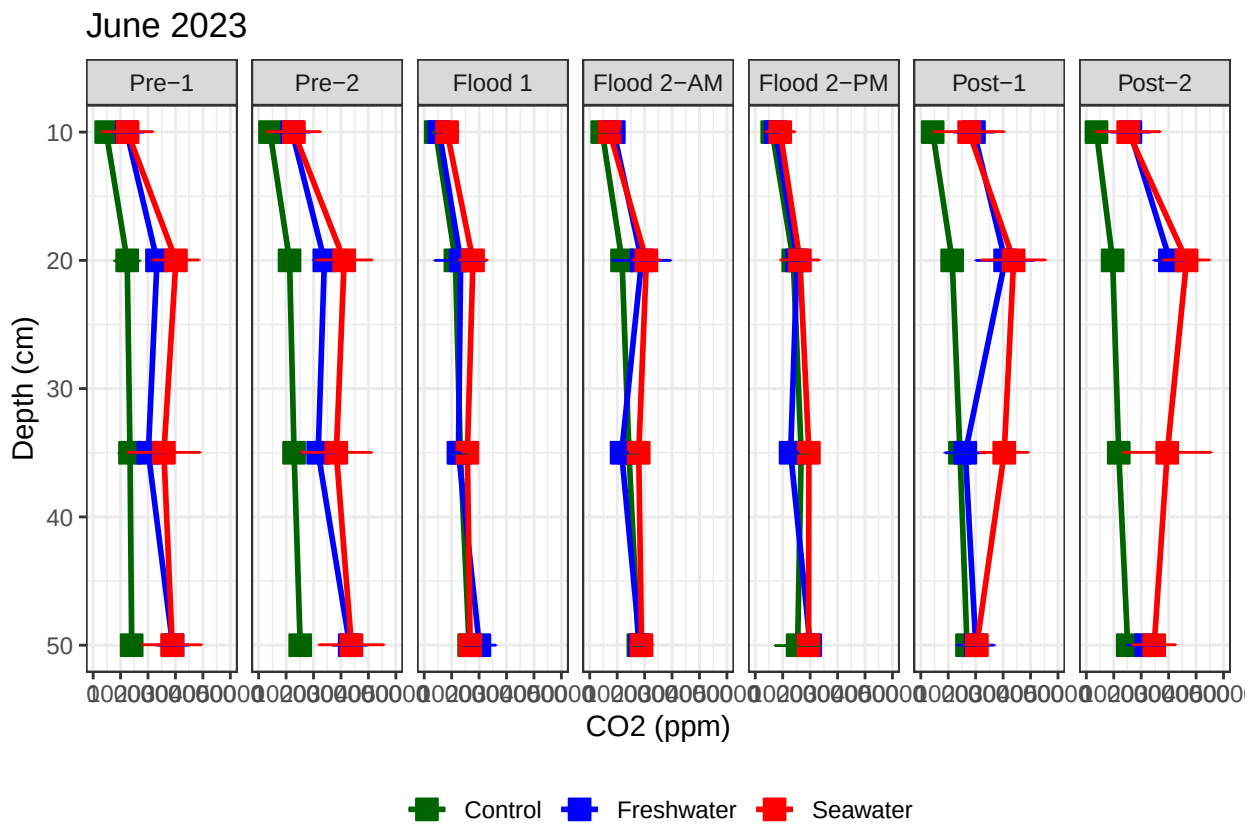
```
  theme_bw() +
  theme(legend.position="bottom") +
  theme(legend.title=element_blank())+
  geom_errorbar(aes(Depth, ymin=mean_CH4_ppm-sd_CH4,
                    ymax=mean_CH4_ppm+sd_CH4),width=0.3,position=position_dodge(.1))
june_avg_soil_ch41
```




```
june_avg_soil_co21 <- ggplot(data=ghg_avg1, aes(col=Sample_Plot, fill=Sample_Plot)) +
  geom_line(size=1, aes(Depth, mean_CO2_ppm)) +
  geom_point(shape=22, size=4, aes(Depth, mean_CO2_ppm)) +
  scale_color_manual(values=c( "darkgreen","blue", "red")) +
  scale_fill_manual(values=c( "darkgreen","blue", "red")) +
  coord_flip() +
  scale_x_reverse() +
  ylim(0,50000) +
  facet_grid(~factor(Event, levels=c("Pre-1", "Pre-2", "Flood 1","Flood 2-AM" ,"Flood 2-PM", "Post-1","P
  labs(title = "June 2023", y="CO2 (ppm)", x="Depth (cm)") +
  theme_bw() +
  theme(legend.position="bottom") +
  theme(legend.title=element_blank())+
  geom_errorbar(aes(Depth, ymin=mean_CO2_ppm-sd_CO2,
                    ymax=mean_CO2_ppm+sd_CO2),width=0.3,position=position_dodge(.1))
june_avg_soil_co21
```

```
## Warning: Removed 1 row containing missing values or values outside the scale range
```

```
## (`geom_point()`).
```




```
##################################################################################################
```

July 2023

```
#### July 2023 ####
##################################################################################################
#read in some data
dat <- read.csv("2023 Processed Data/COMPASS_TEMPEST_SGW_202307_Processed.csv")
head(dat)
```

```
##   X   Machine       User Run_Date Sample_Year Sample_Month Sample_Day
## 1 1 Varian GC Max Wegner 20231205        2023         July         12
## 2 2 Varian GC Max Wegner 20231205        2023         July         12
## 3 3 Varian GC Max Wegner 20231205        2023         July         12
## 4 4 Varian GC Max Wegner 20231205        2023         July         12
## 5 5 Varian GC Max Wegner 20231205        2023         July         12
## 6 6 Varian GC Max Wegner 20231205        2023         July         12
##   Sample_Time Time_Zone G_W Sample_Plot Sample_Grid Sample_Depth Sample_ID
## 1        1151       EDT   G     Control          C3           10         1
## 2        1153       EDT   G     Control          C3           20         2
## 3        1154       EDT   G     Control          C3           35         3
```

```
## 4        1155       EDT   G     Control          C3           50         4
## 5        1156       EDT   G     Control          C6           10         5
## 6        1158       EDT   G     Control          C6           20         6
##   Dilution_Factor Field_Notes CO2_Conc_ppm     CH4_Flag       CO2_Flag
## 1               1          NA      6798.45 Within Range   Within Range
## 2               1          NA     11090.58 Within Range   Within Range
## 3               1          NA     17086.75 Within Range   Within Range
## 4               1          NA     20515.42 Within Range   Within Range
## 5               1          NA      6794.19 Within Range   Within Range
## 6               1          NA     53218.30 Within Range Needs Dilution
##   CH4_Conc_ppm_dilcorr CO2_Conc_ppm_dilcorr CH4_Conc_umol CO2_Conc_umol
## 1            2.4698744              6798.45    153.982193      423843.5
## 2            0.4998265             11090.58     31.161255      691432.4
## 3            0.2773232             17086.75     17.289475     1065258.8
## 4            0.1441460             20515.42      8.986658     1279016.1
## 5            1.0877551              6794.19     67.815154      423577.9
## 6            1.3297478             53218.30     82.901980     3317849.2
```

```r
dat <-  dat %>%
  filter(!str_detect(Sample_Plot, "Transition"))
head(dat)
```

```
##   X   Machine       User Run_Date Sample_Year Sample_Month Sample_Day
## 1 1 Varian GC Max Wegner 20231205        2023         July         12
## 2 2 Varian GC Max Wegner 20231205        2023         July         12
## 3 3 Varian GC Max Wegner 20231205        2023         July         12
## 4 4 Varian GC Max Wegner 20231205        2023         July         12
## 5 5 Varian GC Max Wegner 20231205        2023         July         12
## 6 6 Varian GC Max Wegner 20231205        2023         July         12
##   Sample_Time Time_Zone G_W Sample_Plot Sample_Grid Sample_Depth Sample_ID
## 1        1151       EDT   G     Control          C3           10         1
## 2        1153       EDT   G     Control          C3           20         2
## 3        1154       EDT   G     Control          C3           35         3
## 4        1155       EDT   G     Control          C3           50         4
## 5        1156       EDT   G     Control          C6           10         5
## 6        1158       EDT   G     Control          C6           20         6
##   Dilution_Factor Field_Notes CO2_Conc_ppm     CH4_Flag       CO2_Flag
## 1               1          NA      6798.45 Within Range   Within Range
## 2               1          NA     11090.58 Within Range   Within Range
## 3               1          NA     17086.75 Within Range   Within Range
## 4               1          NA     20515.42 Within Range   Within Range
## 5               1          NA      6794.19 Within Range   Within Range
## 6               1          NA     53218.30 Within Range Needs Dilution
##   CH4_Conc_ppm_dilcorr CO2_Conc_ppm_dilcorr CH4_Conc_umol CO2_Conc_umol
## 1            2.4698744              6798.45    153.982193      423843.5
## 2            0.4998265             11090.58     31.161255      691432.4
## 3            0.2773232             17086.75     17.289475     1065258.8
## 4            0.1441460             20515.42      8.986658     1279016.1
## 5            1.0877551              6794.19     67.815154      423577.9
## 6            1.3297478             53218.30     82.901980     3317849.2
```

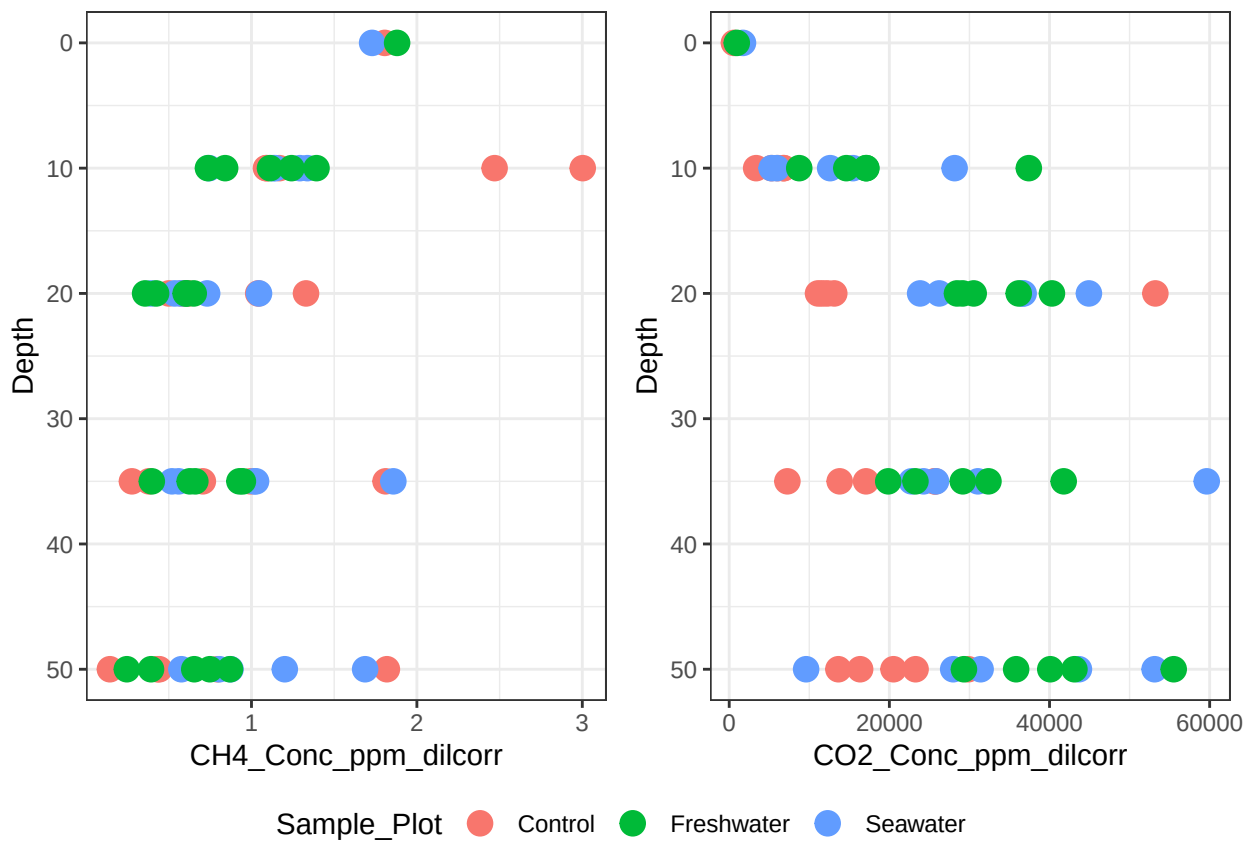
```r
dat$Depth <- dat$Sample_Depth
dat$Depth <- revalue(dat$Depth, c("10" = 10, "20" = 20, "35" = 35, "50"= 50, "ATM" = 0))
```

```r
dat$Depth <- as.numeric(dat$Depth)

soil_ch4 <- ggplot(data=dat, aes( x=CH4_Conc_ppm_dilcorr, y=Depth, color=Sample_Plot)) +
  geom_point(size=4)  + scale_y_reverse() + theme_bw()

soil_co2 <- ggplot(data=dat, aes( x=CO2_Conc_ppm_dilcorr, y=Depth, color=Sample_Plot)) +
  geom_point(size=4)  + scale_y_reverse() + theme_bw()

ggarrange(soil_ch4, soil_co2, nrow=1, ncol=2, common.legend = TRUE, legend="bottom")
```




```r
dat$Sample_Month <- as.factor(dat$Sample_Month)
dat$Sample_Depth <- as.factor(dat$Sample_Depth)
dat$Sample_Plot <- as.factor(dat$Sample_Plot)

levels(dat$Sample_Plot)
```

```
## [1] "Control"    "Freshwater" "Seawater"
```

```r
dat <- as.data.frame(dat)

ghg_avg <- dat %>%
  group_by(Sample_Depth, Sample_Plot) %>%
  dplyr::summarize(mean_CH4_ppm = mean(CH4_Conc_ppm_dilcorr, na.rm=TRUE),
                   sd_CH4 = sd(CH4_Conc_ppm_dilcorr, na.rm=TRUE),
```

```
                    mean_CO2_ppm = mean(CO2_Conc_ppm_dilcorr, na.rm=TRUE),
                    sd_CO2 = sd(CO2_Conc_ppm_dilcorr, na.rm=TRUE))
```

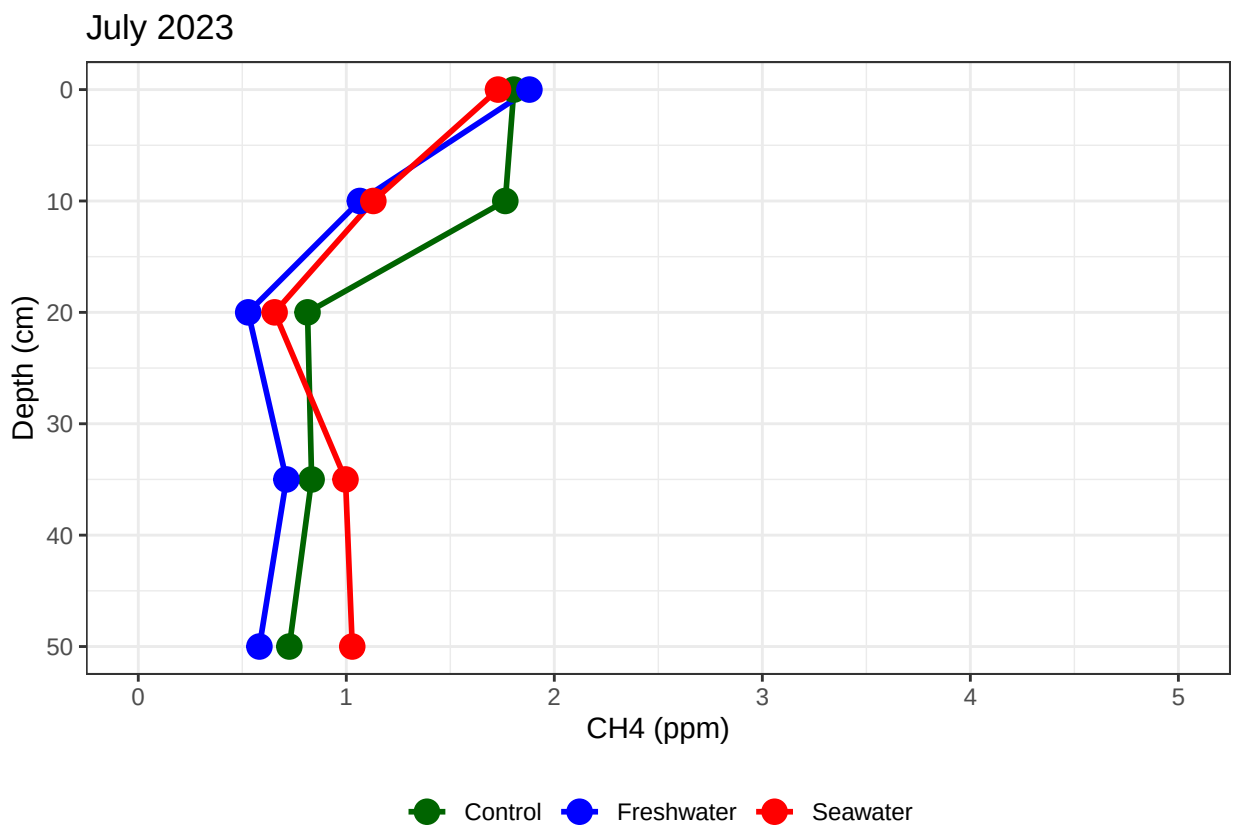
```
## `summarise()` has grouped output by 'Sample_Depth'. You can override using the
## `.groups` argument.
```

```
head(ghg_avg)
```

```
## # A tibble: 6 x 6
## # Groups:   Sample_Depth [2]
##   Sample_Depth Sample_Plot mean_CH4_ppm sd_CH4 mean_CO2_ppm sd_CO2
##   <fct>        <fct>              <dbl>  <dbl>        <dbl>  <dbl>
## 1 10           Control            1.76   0.908        5659.  1427.
## 2 10           Freshwater         1.07   0.273       19001. 10847.
## 3 10           Seawater           1.13   0.234       13490.  9268.
## 4 20           Control            0.814  0.357       20265. 18437.
## 5 20           Freshwater         0.528  0.130       32918.  5121.
## 6 20           Seawater           0.655  0.249       33674.  8615.
```

```
ghg_avg$Depth <- revalue(ghg_avg$Sample_Depth, c("10" = 10, "20" = 20, "35" = 35, "50"= 50, "ATM" = 0))
ghg_avg$Depth <- as.character(ghg_avg$Depth)
ghg_avg$Depth <- as.numeric(ghg_avg$Depth)

july_avg_soil_ch4 <- ggplot(data=ghg_avg, aes(col=Sample_Plot, fill=Sample_Plot)) +
  geom_line(size=1, aes(Depth, mean_CH4_ppm)) +
  geom_point(size=4, aes(Depth, mean_CH4_ppm)) +
  scale_color_manual(values=c( "darkgreen","blue", "red")) +
  scale_fill_manual(values=c( "darkgreen","blue", "red")) +
  coord_flip() +
  scale_x_reverse() +
  ylim(0,5) +
  labs(title = "July 2023", y="CH4 (ppm)", x="Depth (cm)") +
  theme_bw() +
  theme(legend.position="bottom") +
  theme(legend.title=element_blank())
july_avg_soil_ch4
```

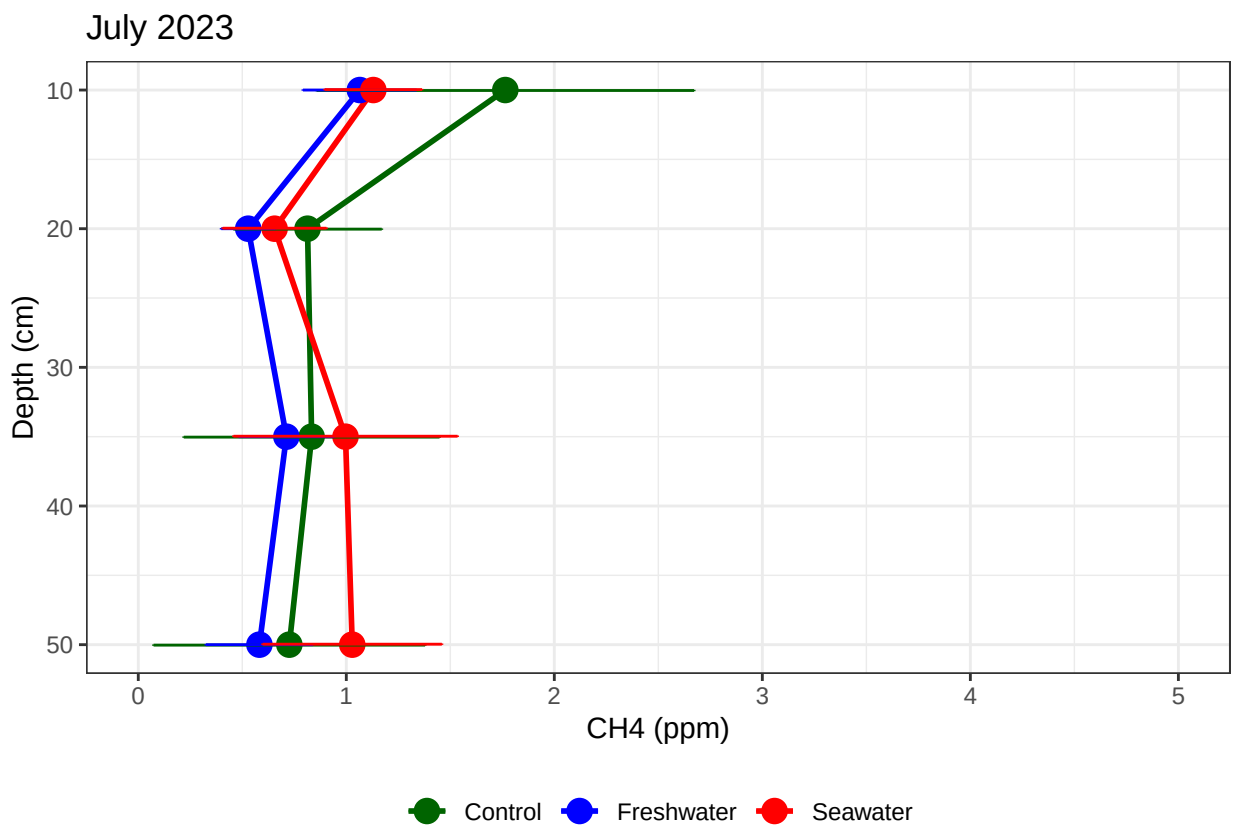



```
ghg_avg1 <-  ghg_avg %>%
  filter(!(Sample_Depth == "ATM"))
head(ghg_avg1)
```

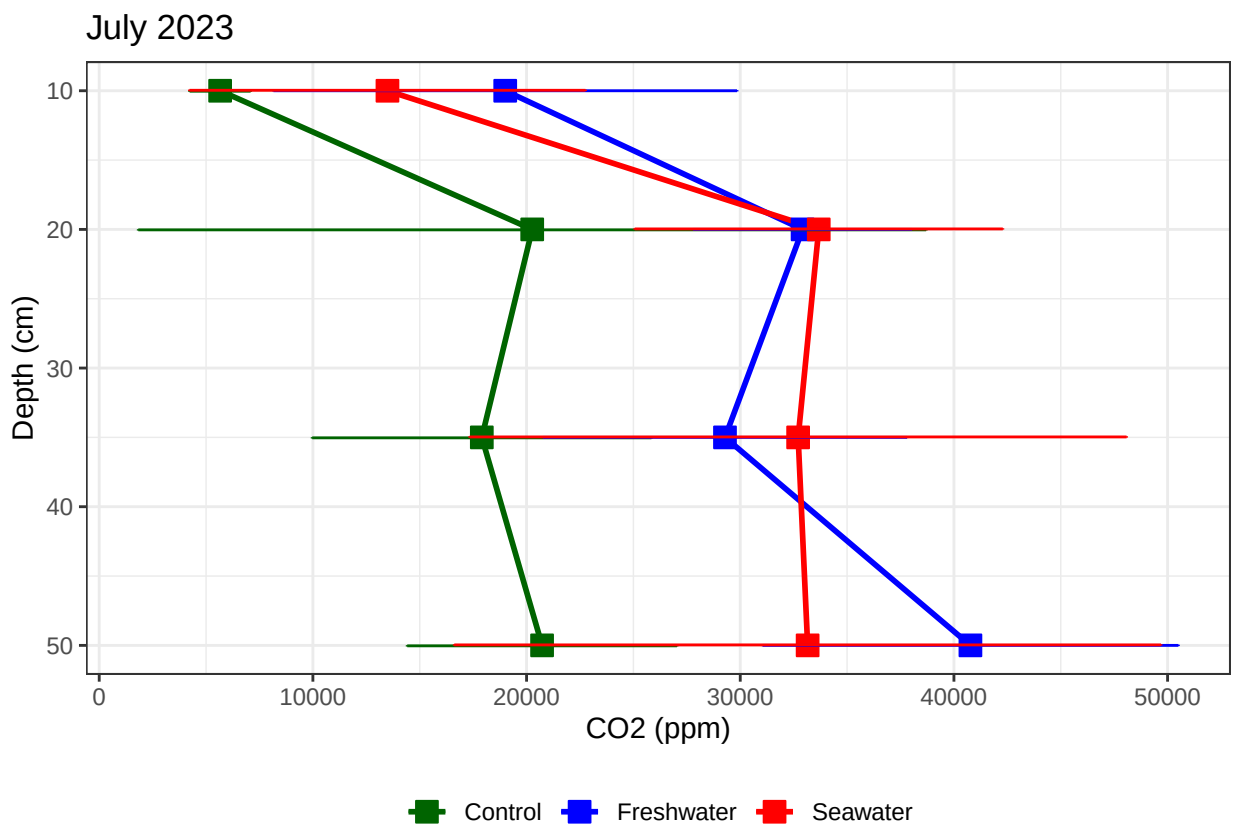
```
## # A tibble: 6 x 7
## # Groups:   Sample_Depth [2]
##   Sample_Depth Sample_Plot mean_CH4_ppm sd_CH4 mean_CO2_ppm sd_CO2 Depth
##   <fct>        <fct>              <dbl>  <dbl>        <dbl>  <dbl> <dbl>
## 1 10           Control            1.76   0.908        5659.  1427.    10
## 2 10           Freshwater         1.07   0.273       19001. 10847.    10
## 3 10           Seawater           1.13   0.234       13490.  9268.    10
## 4 20           Control            0.814  0.357       20265. 18437.    20
## 5 20           Freshwater         0.528  0.130       32918.  5121.    20
## 6 20           Seawater           0.655  0.249       33674.  8615.    20
```

```
july_avg_soil_ch41 <- ggplot(data=ghg_avg1, aes(col=Sample_Plot, fill=Sample_Plot)) +
  geom_line(size=1, aes(Depth, mean_CH4_ppm)) +
  geom_point(size=4, aes(Depth, mean_CH4_ppm)) +
  scale_color_manual(values=c( "darkgreen","blue", "red")) +
  scale_fill_manual(values=c( "darkgreen","blue", "red")) +
  coord_flip() +
  scale_x_reverse() +
  ylim(0,5) +
  labs(title = "July 2023", y="CH4 (ppm)", x="Depth (cm)") +
  theme_bw() +
```

```
  theme(legend.position="bottom") +
  theme(legend.title=element_blank())+
  geom_errorbar(aes(Depth, ymin=mean_CH4_ppm-sd_CH4,
                    ymax=mean_CH4_ppm+sd_CH4),width=0.3,position=position_dodge(.1))
july_avg_soil_ch41
```




```
july_avg_soil_co21 <- ggplot(data=ghg_avg1, aes(col=Sample_Plot, fill=Sample_Plot)) +
  geom_line(size=1, aes(Depth, mean_CO2_ppm)) +
  geom_point(shape=22, size=4, aes(Depth, mean_CO2_ppm)) +
  scale_color_manual(values=c( "darkgreen","blue", "red")) +
  scale_fill_manual(values=c( "darkgreen","blue", "red")) +
  coord_flip() +
  scale_x_reverse() +
  #ylim(0,5) +
  labs(title = "July 2023", y="CO2 (ppm)", x="Depth (cm)") +
  theme_bw() +
  theme(legend.position="bottom") +
  theme(legend.title=element_blank())+
  geom_errorbar(aes(Depth, ymin=mean_CO2_ppm-sd_CO2,
                    ymax=mean_CO2_ppm+sd_CO2),width=0.3,position=position_dodge(.1))
july_avg_soil_co21
```




```
##################################################################################################
```

August 2023

```
#### Aug 2023 ####
##################################################################################################
#read in some data
dat <- read.csv("2023 Processed Data/COMPASS_TEMPEST_SGW_202308_Processed.csv")
head(dat)
```

```
##   X   Machine       User Run_Date Sample_Year Sample_Month Sample_Day
## 1 1 Varian GC Max Wegner 20231211        2023       August         14
## 2 2 Varian GC Max Wegner 20231211        2023       August         14
## 3 3 Varian GC Max Wegner 20231211        2023       August         14
## 4 4 Varian GC Max Wegner 20231211        2023       August         14
## 5 5 Varian GC Max Wegner 20231211        2023       August         14
## 6 6 Varian GC Max Wegner 20231211        2023       August         14
##   Sample_Time Time_Zone G_W Sample_Plot Sample_Grid Sample_Depth Sample_ID
## 1         949       EDT   G     Control          C3           10         1
## 2         950       EDT   G     Control          C3           20         2
## 3         951       EDT   G     Control          C3           35         3
## 4         952       EDT   G     Control          C3           50         4
## 5         955       EDT   G     Control          C6           10         5
```

```
## 6         956       EDT   G     Control          C6           20         6
##   Dilution_Factor Field_Notes CO2_Conc_ppm     CH4_Flag     CO2_Flag
## 1               1                 6210.859 Within Range Within Range
## 2               1                11688.642 Within Range Within Range
## 3               1                14556.794 Within Range Within Range
## 4               1                26999.751 Within Range Within Range
## 5               1                 9538.098 Within Range Within Range
## 6               1                29941.999 Within Range Within Range
##   CH4_Conc_ppm_dilcorr CO2_Conc_ppm_dilcorr CH4_Conc_umol CO2_Conc_umol
## 1            0.7275841             6210.859      45.36060      387210.7
## 2            0.7784276            11688.642      48.53040      728718.3
## 3            0.6574551            14556.794      40.98847      907530.8
## 4            0.4873921            26999.751      30.38604     1683276.2
## 5            1.7970519             9538.098     112.03565      594644.5
## 6            1.1764099            29941.999      73.34226     1866708.1
```

```
dat <-  dat %>%
  filter(!str_detect(Sample_Plot, "Transition"))
head(dat)
```

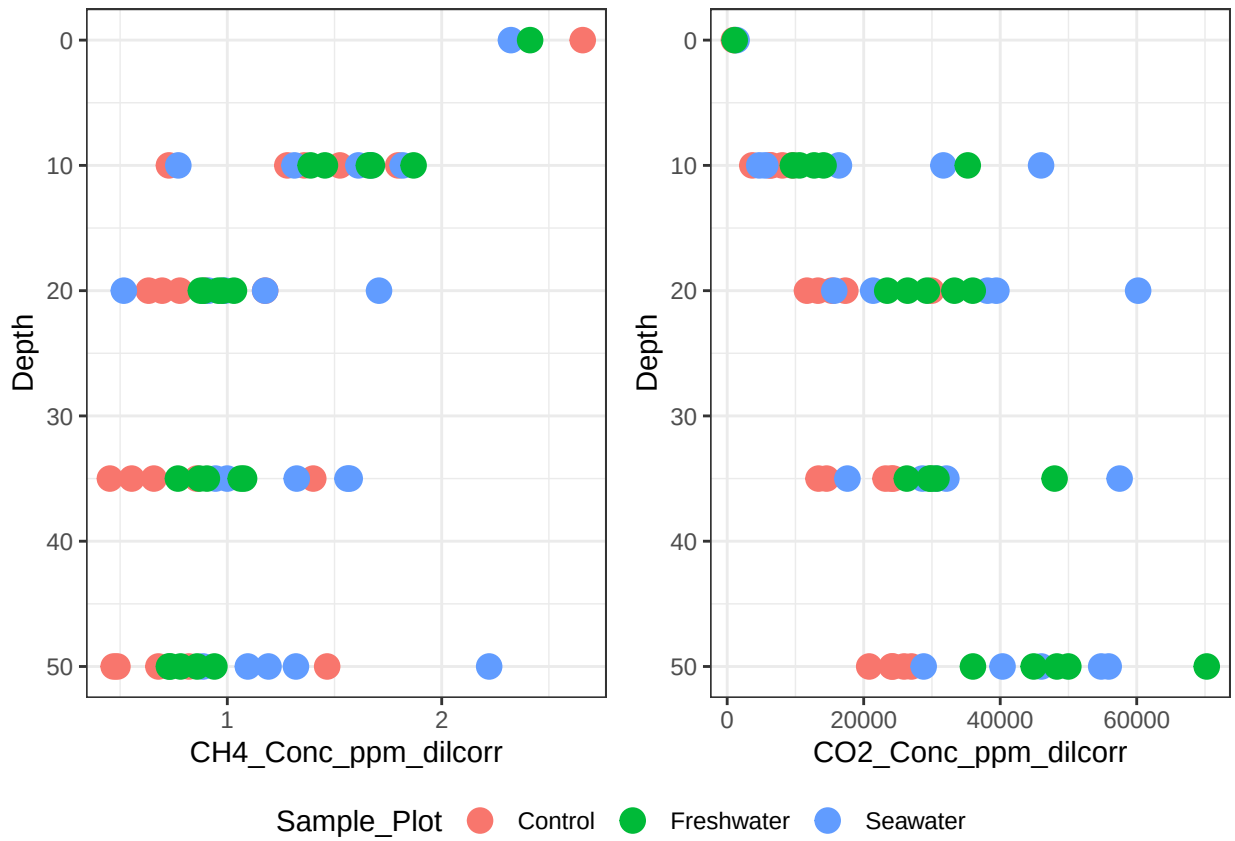
```
##   X   Machine       User Run_Date Sample_Year Sample_Month Sample_Day
## 1 1 Varian GC Max Wegner 20231211        2023       August         14
## 2 2 Varian GC Max Wegner 20231211        2023       August         14
## 3 3 Varian GC Max Wegner 20231211        2023       August         14
## 4 4 Varian GC Max Wegner 20231211        2023       August         14
## 5 5 Varian GC Max Wegner 20231211        2023       August         14
## 6 6 Varian GC Max Wegner 20231211        2023       August         14
##   Sample_Time Time_Zone G_W Sample_Plot Sample_Grid Sample_Depth Sample_ID
## 1         949       EDT   G     Control          C3           10         1
## 2         950       EDT   G     Control          C3           20         2
## 3         951       EDT   G     Control          C3           35         3
## 4         952       EDT   G     Control          C3           50         4
## 5         955       EDT   G     Control          C6           10         5
## 6         956       EDT   G     Control          C6           20         6
##   Dilution_Factor Field_Notes CO2_Conc_ppm     CH4_Flag     CO2_Flag
## 1               1                 6210.859 Within Range Within Range
## 2               1                11688.642 Within Range Within Range
## 3               1                14556.794 Within Range Within Range
## 4               1                26999.751 Within Range Within Range
## 5               1                 9538.098 Within Range Within Range
## 6               1                29941.999 Within Range Within Range
##   CH4_Conc_ppm_dilcorr CO2_Conc_ppm_dilcorr CH4_Conc_umol CO2_Conc_umol
## 1            0.7275841             6210.859      45.36060      387210.7
## 2            0.7784276            11688.642      48.53040      728718.3
## 3            0.6574551            14556.794      40.98847      907530.8
## 4            0.4873921            26999.751      30.38604     1683276.2
## 5            1.7970519             9538.098     112.03565      594644.5
## 6            1.1764099            29941.999      73.34226     1866708.1
```

```
dat$Depth <- dat$Sample_Depth
dat$Depth <- revalue(dat$Depth, c("10" = 10, "20" = 20, "35" = 35, "50"= 50, "ATM" = 0))
dat$Depth <- as.numeric(dat$Depth)
```

```r
soil_ch4 <- ggplot(data=dat, aes( x=CH4_Conc_ppm_dilcorr, y=Depth, color=Sample_Plot)) +
  geom_point(size=4)  + scale_y_reverse() + theme_bw()

soil_co2 <- ggplot(data=dat, aes( x=CO2_Conc_ppm_dilcorr, y=Depth, color=Sample_Plot)) +
  geom_point(size=4)  + scale_y_reverse() + theme_bw()

ggarrange(soil_ch4, soil_co2, nrow=1, ncol=2, common.legend = TRUE, legend="bottom")
```




```r
dat$Sample_Month <- as.factor(dat$Sample_Month)
dat$Sample_Depth <- as.factor(dat$Sample_Depth)
dat$Sample_Plot <- as.factor(dat$Sample_Plot)

levels(dat$Sample_Plot)
```

```
## [1] "Control"    "Freshwater" "Seawater"
```

```r
dat <- as.data.frame(dat)

ghg_avg <- dat %>%
  group_by(Sample_Depth, Sample_Plot) %>%
  dplyr::summarize(mean_CH4_ppm = mean(CH4_Conc_ppm_dilcorr, na.rm=TRUE),
                   sd_CH4 = sd(CH4_Conc_ppm_dilcorr, na.rm=TRUE),
                   mean_CO2_ppm = mean(CO2_Conc_ppm_dilcorr, na.rm=TRUE),
                   sd_CO2 = sd(CO2_Conc_ppm_dilcorr, na.rm=TRUE))
```

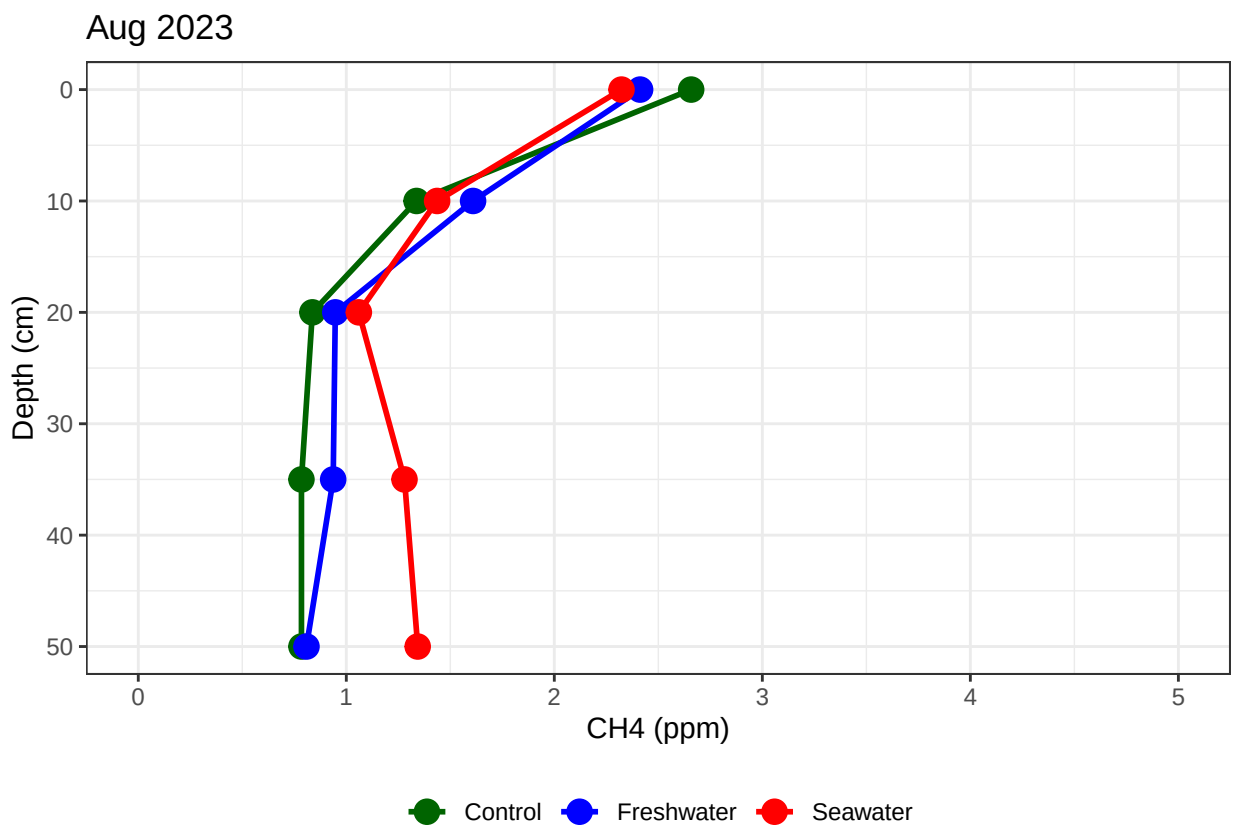
```
## `summarise()` has grouped output by 'Sample_Depth'. You can override using the
## `.groups` argument.
```

```r
head(ghg_avg)
```

```
## # A tibble: 6 x 6
## # Groups:   Sample_Depth [2]
##   Sample_Depth Sample_Plot mean_CH4_ppm sd_CH4 mean_CO2_ppm sd_CO2
##   <fct>        <fct>              <dbl>  <dbl>        <dbl>  <dbl>
## 1 10           Control            1.34  0.394         6792.  2198.
## 2 10           Freshwater         1.61  0.192        16464. 10646.
## 3 10           Seawater           1.44  0.415        20859. 17773.
## 4 20           Control            0.837 0.215        17515.  7262.
## 5 20           Freshwater         0.947 0.0638       29684.  5055.
## 6 20           Seawater           1.06  0.434        34970. 17487.
```

```r
ghg_avg$Depth <- revalue(ghg_avg$Sample_Depth, c("10" = 10, "20" = 20, "35" = 35, "50"= 50, "ATM" = 0))
ghg_avg$Depth <- as.character(ghg_avg$Depth)
ghg_avg$Depth <- as.numeric(ghg_avg$Depth)

aug_avg_soil_ch4 <- ggplot(data=ghg_avg, aes(col=Sample_Plot, fill=Sample_Plot)) +
  geom_line(size=1, aes(Depth, mean_CH4_ppm)) +
  geom_point(size=4, aes(Depth, mean_CH4_ppm)) +
  scale_color_manual(values=c( "darkgreen","blue", "red")) +
  scale_fill_manual(values=c( "darkgreen","blue", "red")) +
  coord_flip() +
  scale_x_reverse() +
  ylim(0,5) +
  labs(title = "Aug 2023", y="CH4 (ppm)", x="Depth (cm)") +
  theme_bw() +
  theme(legend.position="bottom") +
  theme(legend.title=element_blank())
aug_avg_soil_ch4
```

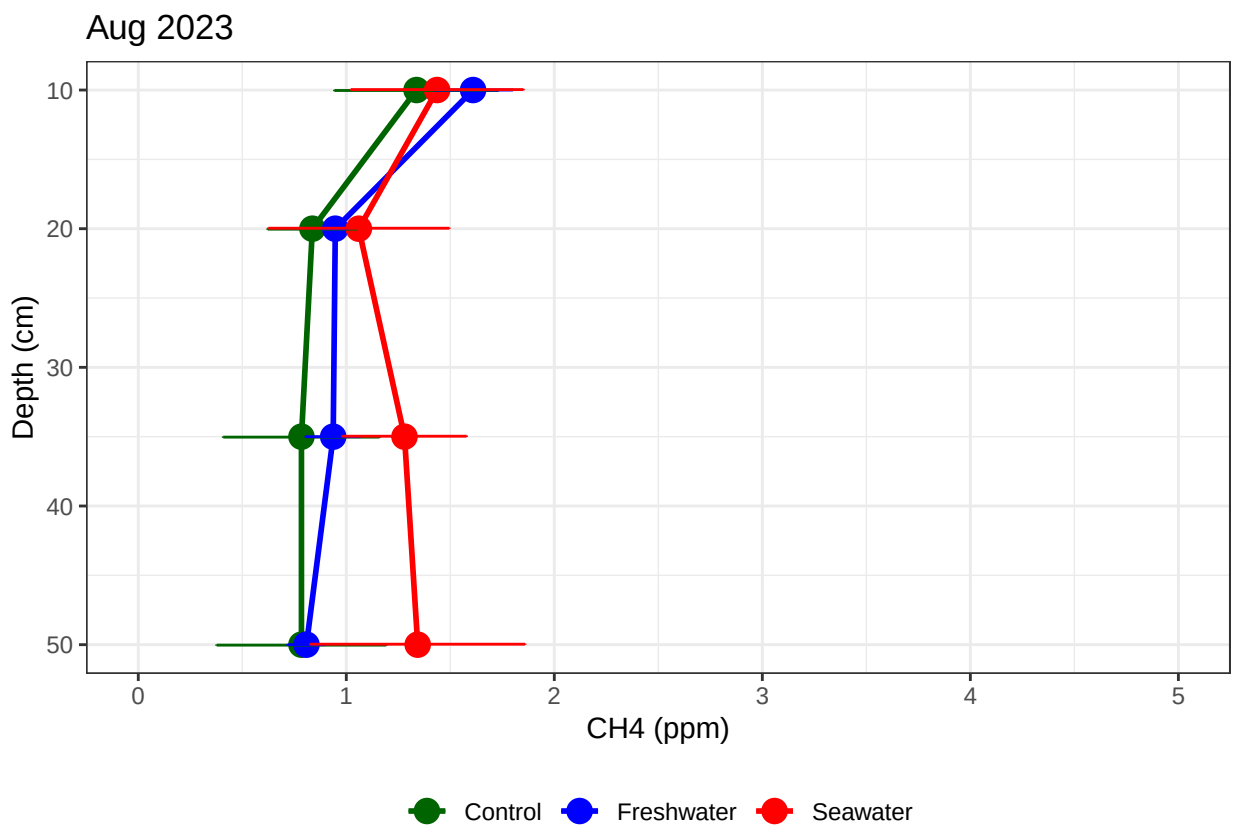



```
ghg_avg1 <-  ghg_avg %>%
  filter(!(Sample_Depth == "ATM"))
head(ghg_avg1)
```

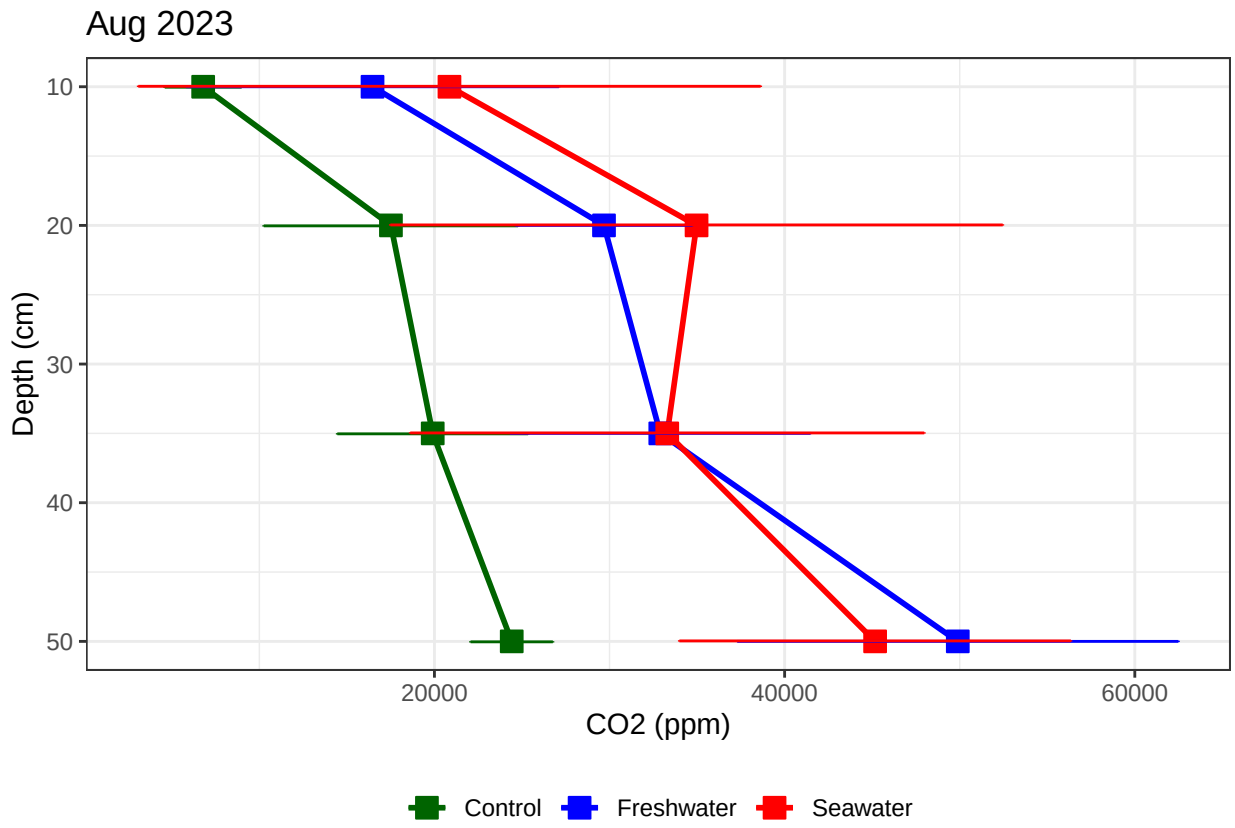
```
## # A tibble: 6 x 7
## # Groups:   Sample_Depth [2]
##   Sample_Depth Sample_Plot mean_CH4_ppm sd_CH4 mean_CO2_ppm sd_CO2 Depth
##   <fct>        <fct>              <dbl>  <dbl>        <dbl>  <dbl> <dbl>
## 1 10           Control            1.34  0.394         6792.  2198.    10
## 2 10           Freshwater         1.61  0.192        16464. 10646.    10
## 3 10           Seawater           1.44  0.415        20859. 17773.    10
## 4 20           Control            0.837 0.215        17515.  7262.    20
## 5 20           Freshwater         0.947 0.0638       29684.  5055.    20
## 6 20           Seawater           1.06  0.434        34970. 17487.    20
```

```
aug_avg_soil_ch41 <- ggplot(data=ghg_avg1, aes(col=Sample_Plot, fill=Sample_Plot)) +
  geom_line(size=1, aes(Depth, mean_CH4_ppm)) +
  geom_point(size=4, aes(Depth, mean_CH4_ppm)) +
  scale_color_manual(values=c( "darkgreen","blue", "red")) +
  scale_fill_manual(values=c( "darkgreen","blue", "red")) +
  coord_flip() +
  scale_x_reverse() +
  ylim(0,5) +
  labs(title = "Aug 2023", y="CH4 (ppm)", x="Depth (cm)") +
  theme_bw() +
```

```
  theme(legend.position="bottom") +
  theme(legend.title=element_blank())+
  geom_errorbar(aes(Depth, ymin=mean_CH4_ppm-sd_CH4,
                    ymax=mean_CH4_ppm+sd_CH4),width=0.3,position=position_dodge(.1))
aug_avg_soil_ch41
```




```
aug_avg_soil_co21 <- ggplot(data=ghg_avg1, aes(col=Sample_Plot, fill=Sample_Plot)) +
  geom_line(size=1, aes(Depth, mean_CO2_ppm)) +
  geom_point(shape=22, size=4, aes(Depth, mean_CO2_ppm)) +
  scale_color_manual(values=c( "darkgreen","blue", "red")) +
  scale_fill_manual(values=c( "darkgreen","blue", "red")) +
  coord_flip() +
  scale_x_reverse() +
  #ylim(0,5) +
  labs(title = "Aug 2023", y="CO2 (ppm)", x="Depth (cm)") +
  theme_bw() +
  theme(legend.position="bottom") +
  theme(legend.title=element_blank())+
  geom_errorbar(aes(Depth, ymin=mean_CO2_ppm-sd_CO2,
                    ymax=mean_CO2_ppm+sd_CO2),width=0.3,position=position_dodge(.1))
aug_avg_soil_co21
```




```
##########################################################################################
```

September 2023

```
#### Sept 2023 ####
##########################################################################################
#read in some data
dat <- read.csv("2023 Processed Data/COMPASS_TEMPEST_SGW_202309_Processed.csv")
head(dat)
```

```
##   X   Machine       User Run_Date Sample_Year Sample_Month Sample_Day
## 1 1 Varian GC Max Wegner 20231212        2023    September         12
## 2 2 Varian GC Max Wegner 20231212        2023    September         12
## 3 3 Varian GC Max Wegner 20231212        2023    September         12
## 4 4 Varian GC Max Wegner 20231212        2023    September         12
## 5 5 Varian GC Max Wegner 20231212        2023    September         12
## 6 6 Varian GC Max Wegner 20231212        2023    September         12
##   Sample_Time Time_Zone G_W Sample_Plot Sample_Grid Sample_Depth Sample_ID
## 1        1016       EDT   G     Control          C3           10         1
## 2        1017       EDT   G     Control          C3           20         2
## 3        1018       EDT   G     Control          C3           35         3
## 4        1019       EDT   G     Control          C3           50         4
## 5        1020       EDT   G     Control          C6           10         5
```

```
## 6        1021       EDT   G     Control          C6           20         6
##   Dilution_Factor Field_Notes CO2_Conc_ppm     CH4_Flag     CO2_Flag
## 1               1          NA     4073.666 Within Range Within Range
## 2               1          NA     7476.975 Within Range Within Range
## 3               1          NA    10506.212 Within Range Within Range
## 4               1          NA    16210.371 Within Range Within Range
## 5               1          NA     7861.988 Within Range Within Range
## 6               1          NA    15075.211 Within Range Within Range
##   CH4_Conc_ppm_dilcorr CO2_Conc_ppm_dilcorr CH4_Conc_umol CO2_Conc_umol
## 1            1.4118077             4073.666      88.01793      253969.2
## 2            0.9351395             7476.975      58.30047      466145.6
## 3            0.7698713            10506.212      47.99697      655000.7
## 4            0.5872065            16210.371      36.60889     1010621.6
## 5            1.6205675             7861.988     101.03288      490148.9
## 6            1.4065887            15075.211      87.69256      939851.0
```

```
dat <-  dat %>%
  filter(!str_detect(Sample_Plot, "Transition"))
head(dat)
```

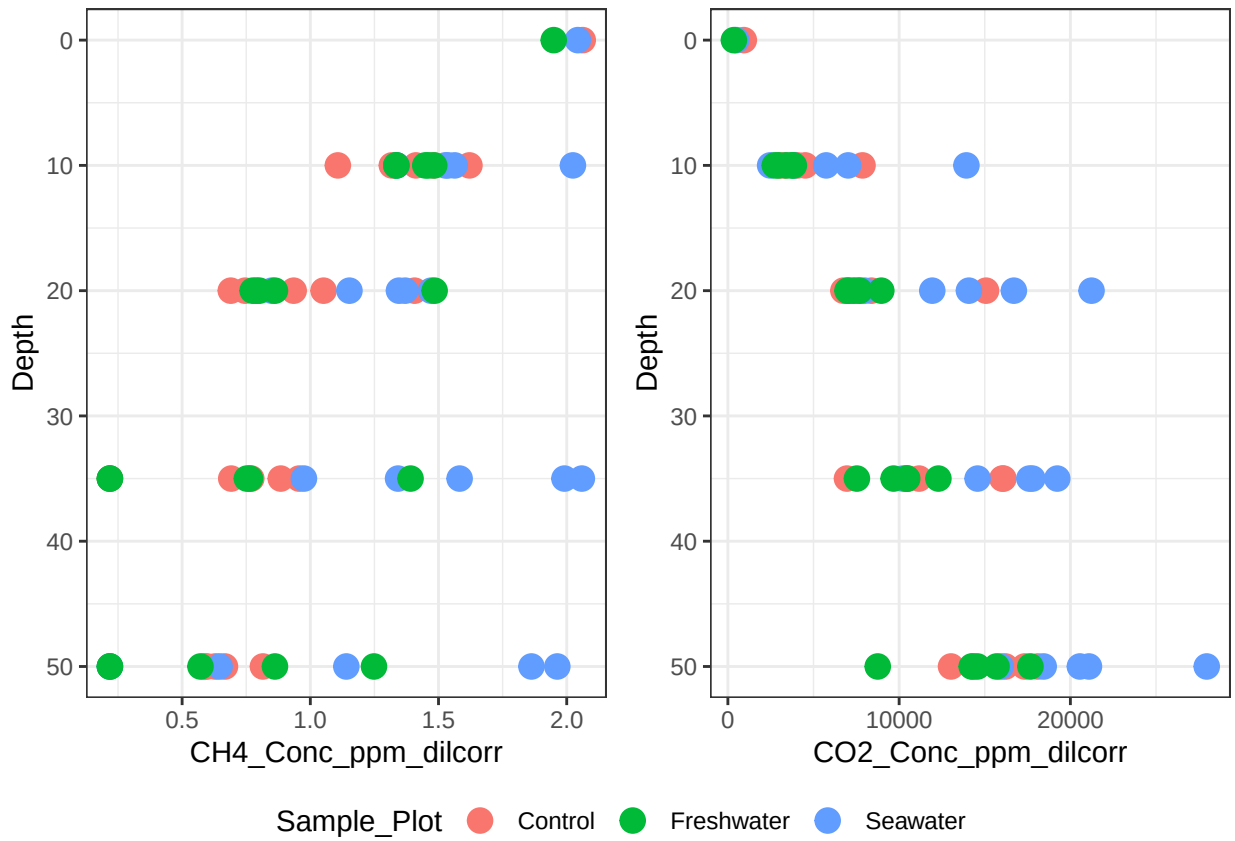
```
##   X   Machine       User Run_Date Sample_Year Sample_Month Sample_Day
## 1 1 Varian GC Max Wegner 20231212        2023    September         12
## 2 2 Varian GC Max Wegner 20231212        2023    September         12
## 3 3 Varian GC Max Wegner 20231212        2023    September         12
## 4 4 Varian GC Max Wegner 20231212        2023    September         12
## 5 5 Varian GC Max Wegner 20231212        2023    September         12
## 6 6 Varian GC Max Wegner 20231212        2023    September         12
##   Sample_Time Time_Zone G_W Sample_Plot Sample_Grid Sample_Depth Sample_ID
## 1        1016       EDT   G     Control          C3           10         1
## 2        1017       EDT   G     Control          C3           20         2
## 3        1018       EDT   G     Control          C3           35         3
## 4        1019       EDT   G     Control          C3           50         4
## 5        1020       EDT   G     Control          C6           10         5
## 6        1021       EDT   G     Control          C6           20         6
##   Dilution_Factor Field_Notes CO2_Conc_ppm     CH4_Flag     CO2_Flag
## 1               1          NA     4073.666 Within Range Within Range
## 2               1          NA     7476.975 Within Range Within Range
## 3               1          NA    10506.212 Within Range Within Range
## 4               1          NA    16210.371 Within Range Within Range
## 5               1          NA     7861.988 Within Range Within Range
## 6               1          NA    15075.211 Within Range Within Range
##   CH4_Conc_ppm_dilcorr CO2_Conc_ppm_dilcorr CH4_Conc_umol CO2_Conc_umol
## 1            1.4118077             4073.666      88.01793      253969.2
## 2            0.9351395             7476.975      58.30047      466145.6
## 3            0.7698713            10506.212      47.99697      655000.7
## 4            0.5872065            16210.371      36.60889     1010621.6
## 5            1.6205675             7861.988     101.03288      490148.9
## 6            1.4065887            15075.211      87.69256      939851.0
```

```
dat$Depth <- dat$Sample_Depth
dat$Depth <- revalue(dat$Depth, c("10" = 10, "20" = 20, "35" = 35, "50"= 50, "ATM" = 0))
dat$Depth <- as.numeric(dat$Depth)
```

```r
soil_ch4 <- ggplot(data=dat, aes( x=CH4_Conc_ppm_dilcorr, y=Depth, color=Sample_Plot)) +
  geom_point(size=4)  + scale_y_reverse() + theme_bw()

soil_co2 <- ggplot(data=dat, aes( x=CO2_Conc_ppm_dilcorr, y=Depth, color=Sample_Plot)) +
  geom_point(size=4)  + scale_y_reverse() + theme_bw()

ggarrange(soil_ch4, soil_co2, nrow=1, ncol=2, common.legend = TRUE, legend="bottom")
```




```r
dat$Sample_Month <- as.factor(dat$Sample_Month)
dat$Sample_Depth <- as.factor(dat$Sample_Depth)
dat$Sample_Plot <- as.factor(dat$Sample_Plot)

levels(dat$Sample_Plot)
```

```
## [1] "Control"    "Freshwater" "Seawater"
```

```r
dat <- as.data.frame(dat)

ghg_avg <- dat %>%
  group_by(Sample_Depth, Sample_Plot) %>%
  dplyr::summarize(mean_CH4_ppm = mean(CH4_Conc_ppm_dilcorr, na.rm=TRUE),
                   sd_CH4 = sd(CH4_Conc_ppm_dilcorr, na.rm=TRUE),
                   mean_CO2_ppm = mean(CO2_Conc_ppm_dilcorr, na.rm=TRUE),
                   sd_CO2 = sd(CO2_Conc_ppm_dilcorr, na.rm=TRUE))
```

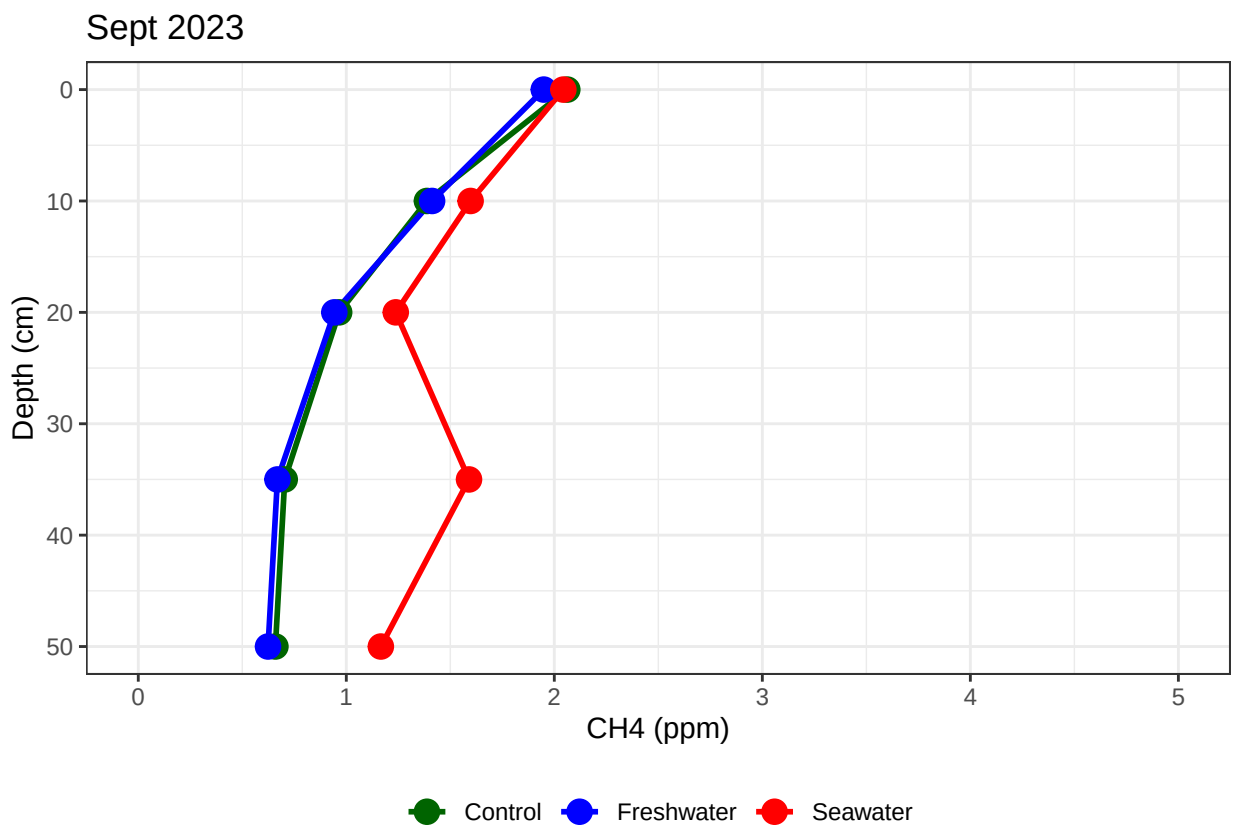
```
## `summarise()` has grouped output by 'Sample_Depth'. You can override using the
## `.groups` argument.
```

```r
head(ghg_avg)
```

```
## # A tibble: 6 x 6
## # Groups:   Sample_Depth [2]
##   Sample_Depth Sample_Plot mean_CH4_ppm sd_CH4 mean_CO2_ppm sd_CO2
##   <fct>        <fct>              <dbl>  <dbl>        <dbl>  <dbl>
## 1 10           Control            1.39  0.192         4564.  1936.
## 2 10           Freshwater         1.41  0.0716        3332.   474.
## 3 10           Seawater           1.60  0.255         6601.  4453.
## 4 20           Control            0.966 0.286         9066.  3409.
## 5 20           Freshwater         0.942 0.305         7601.   810.
## 6 20           Seawater           1.24  0.246        14386.  4985.
```

```r
ghg_avg$Depth <- revalue(ghg_avg$Sample_Depth, c("10" = 10, "20" = 20, "35" = 35, "50"= 50, "ATM" = 0))
ghg_avg$Depth <- as.character(ghg_avg$Depth)
ghg_avg$Depth <- as.numeric(ghg_avg$Depth)

sept_avg_soil_ch4 <- ggplot(data=ghg_avg, aes(col=Sample_Plot, fill=Sample_Plot)) +
  geom_line(size=1, aes(Depth, mean_CH4_ppm)) +
  geom_point(size=4, aes(Depth, mean_CH4_ppm)) +
  scale_color_manual(values=c( "darkgreen","blue", "red")) +
  scale_fill_manual(values=c( "darkgreen","blue", "red")) +
  coord_flip() +
  scale_x_reverse() +
  ylim(0,5) +
  labs(title = "Sept 2023", y="CH4 (ppm)", x="Depth (cm)") +
  theme_bw() +
  theme(legend.position="bottom") +
  theme(legend.title=element_blank())
sept_avg_soil_ch4
```

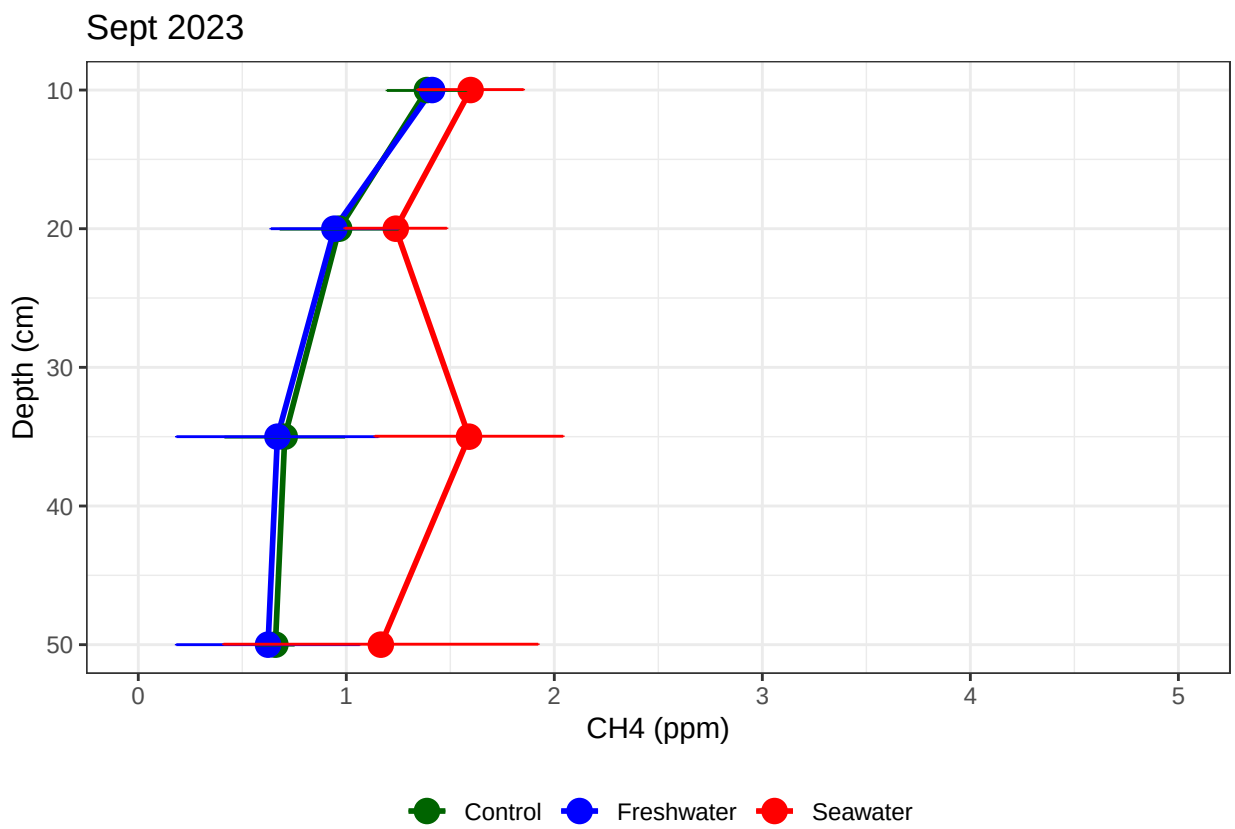



```
ghg_avg1 <-  ghg_avg %>%
  filter(!(Sample_Depth == "ATM"))
head(ghg_avg1)
```

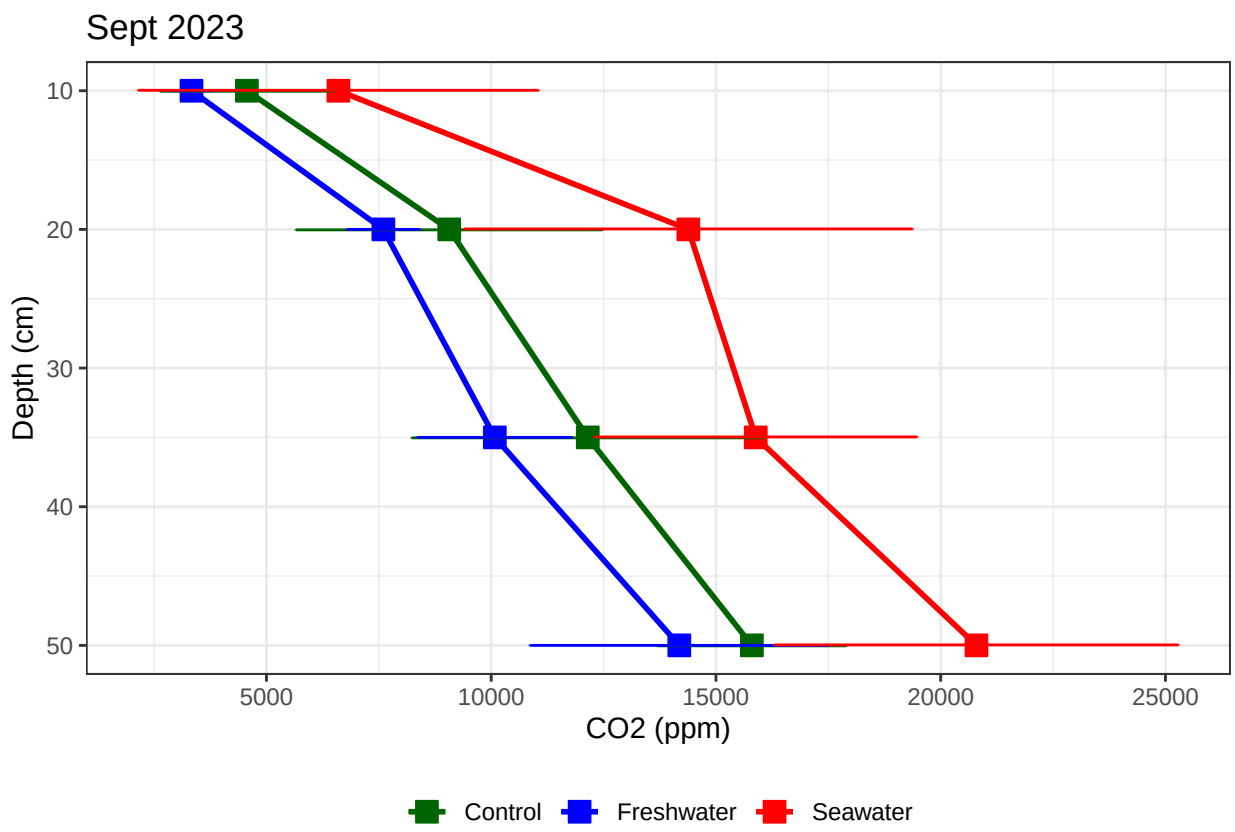
```
## # A tibble: 6 x 7
## # Groups:   Sample_Depth [2]
##   Sample_Depth Sample_Plot mean_CH4_ppm sd_CH4 mean_CO2_ppm sd_CO2 Depth
##   <fct>        <fct>              <dbl>  <dbl>        <dbl>  <dbl> <dbl>
## 1 10           Control            1.39  0.192         4564.  1936.    10
## 2 10           Freshwater         1.41  0.0716        3332.   474.    10
## 3 10           Seawater           1.60  0.255         6601.  4453.    10
## 4 20           Control            0.966 0.286         9066.  3409.    20
## 5 20           Freshwater         0.942 0.305         7601.   810.    20
## 6 20           Seawater           1.24  0.246        14386.  4985.    20
```

```
sept_avg_soil_ch41 <- ggplot(data=ghg_avg1, aes(col=Sample_Plot, fill=Sample_Plot)) +
  geom_line(size=1, aes(Depth, mean_CH4_ppm)) +
  geom_point(size=4, aes(Depth, mean_CH4_ppm)) +
  scale_color_manual(values=c( "darkgreen","blue", "red")) +
  scale_fill_manual(values=c( "darkgreen","blue", "red")) +
  coord_flip() +
  scale_x_reverse() +
  ylim(0,5) +
  labs(title = "Sept 2023", y="CH4 (ppm)", x="Depth (cm)") +
  theme_bw() +
```

```
  theme(legend.position="bottom") +
  theme(legend.title=element_blank())+
  geom_errorbar(aes(Depth, ymin=mean_CH4_ppm-sd_CH4,
                    ymax=mean_CH4_ppm+sd_CH4),width=0.3,position=position_dodge(.1))
sept_avg_soil_ch41
```




```
sept_avg_soil_co21 <- ggplot(data=ghg_avg1, aes(col=Sample_Plot, fill=Sample_Plot)) +
  geom_line(size=1, aes(Depth, mean_CO2_ppm)) +
  geom_point(shape=22, size=4, aes(Depth, mean_CO2_ppm)) +
  scale_color_manual(values=c( "darkgreen","blue", "red")) +
  scale_fill_manual(values=c( "darkgreen","blue", "red")) +
  coord_flip() +
  scale_x_reverse() +
  #ylim(0,5) +
  labs(title = "Sept 2023", y="CO2 (ppm)", x="Depth (cm)") +
  theme_bw() +
  theme(legend.position="bottom") +
  theme(legend.title=element_blank())+
  geom_errorbar(aes(Depth, ymin=mean_CO2_ppm-sd_CO2,
                    ymax=mean_CO2_ppm+sd_CO2),width=0.3,position=position_dodge(.1))
sept_avg_soil_co21
```




```
##################################################################################################
```

October 2023

```
#### OCT 2023 ####
##################################################################################################
#read in some data
dat <- read.csv("2023 Processed Data/COMPASS_TEMPEST_SGW_202310_Processed.csv")
head(dat)
```

```
##   X   Machine       User Run_Date Sample_Year Sample_Month Sample_Day
## 1 1 Varian GC Max Wegner 20240102        2023      October          3
## 2 2 Varian GC Max Wegner 20240102        2023      October          3
## 3 3 Varian GC Max Wegner 20240102        2023      October          3
## 4 4 Varian GC Max Wegner 20240102        2023      October          3
## 5 5 Varian GC Max Wegner 20240102        2023      October          3
## 6 6 Varian GC Max Wegner 20240102        2023      October          3
##   Sample_Time Time_Zone G_W Sample_Plot Sample_Grid Sample_Depth Sample_ID
## 1        1024       EDT   G     Control          C3           10         1
## 2        1025       EDT   G     Control          C3           20         2
## 3        1027       EDT   G     Control          C3           35         3
## 4        1028       EDT   G     Control          C3           50         4
## 5        1019       EDT   G     Control          C6           10         5
```

```
## 6        1020       EDT   G     Control          C6           20         6
##   Dilution_Factor Field_.Notes CO2_Conc_ppm     CH4_Flag     CO2_Flag
## 1               1           NA     3199.083 Within Range Within Range
## 2               1           NA     5913.915 Within Range Within Range
## 3               1           NA    10094.383 Within Range Within Range
## 4               1           NA    15507.908 Within Range Within Range
## 5               1           NA     3668.013 Within Range Within Range
## 6               1           NA    13589.571 Within Range Within Range
##   CH4_Conc_ppm_dilcorr CO2_Conc_ppm_dilcorr CH4_Conc_umol CO2_Conc_umol
## 1             2.532985             3199.083     157.91680      199444.1
## 2             1.917852             5913.915     119.56684      368698.0
## 3             1.648335            10094.383     102.76403      629325.6
## 4             1.519918            15507.908      94.75798      966827.2
## 5             2.713720             3668.013     169.18457      228679.1
## 6             2.331640            13589.571     145.36411      847230.1
```

```r
dat <-  dat %>%
  filter(!str_detect(Sample_Plot, "Transition"))
head(dat)
```

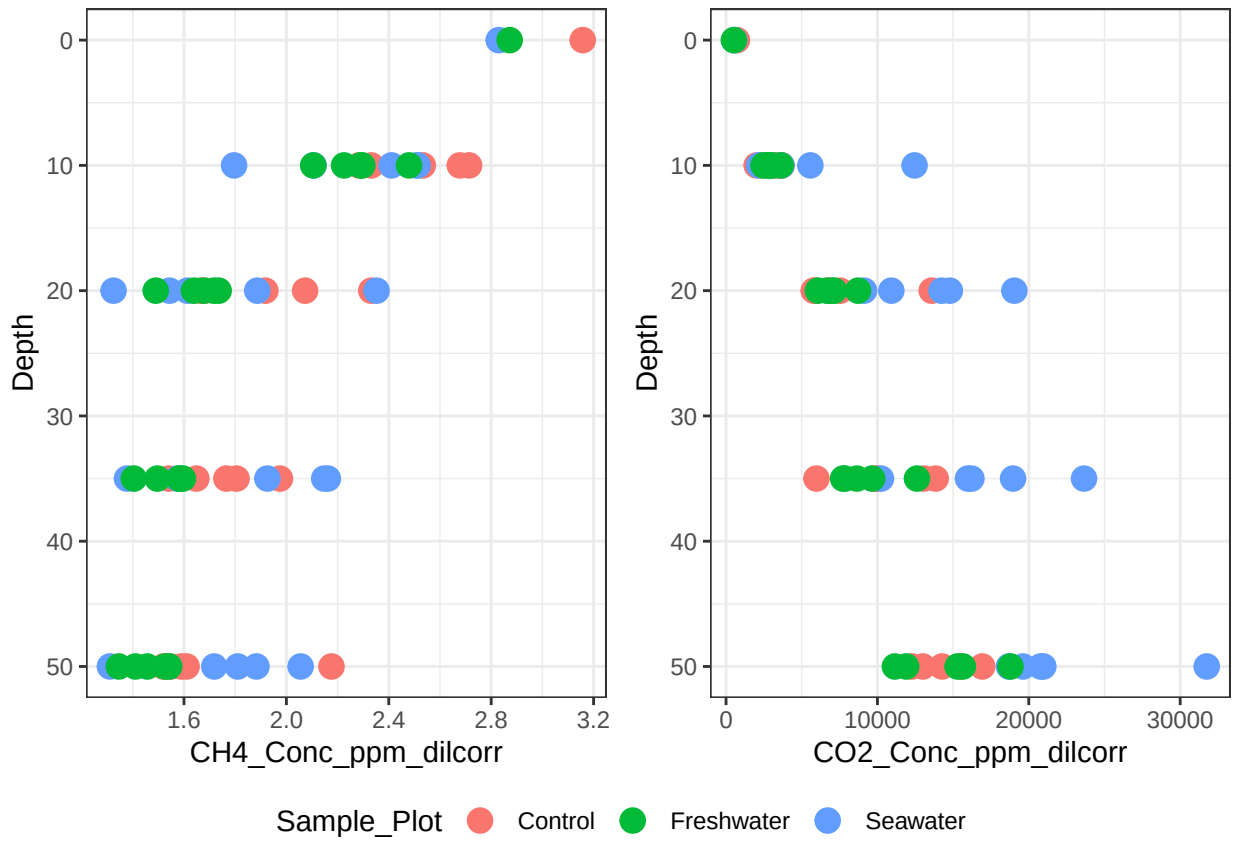
```
##   X   Machine       User Run_Date Sample_Year Sample_Month Sample_Day
## 1 1 Varian GC Max Wegner 20240102        2023      October          3
## 2 2 Varian GC Max Wegner 20240102        2023      October          3
## 3 3 Varian GC Max Wegner 20240102        2023      October          3
## 4 4 Varian GC Max Wegner 20240102        2023      October          3
## 5 5 Varian GC Max Wegner 20240102        2023      October          3
## 6 6 Varian GC Max Wegner 20240102        2023      October          3
##   Sample_Time Time_Zone G_W Sample_Plot Sample_Grid Sample_Depth Sample_ID
## 1        1024       EDT   G     Control          C3           10         1
## 2        1025       EDT   G     Control          C3           20         2
## 3        1027       EDT   G     Control          C3           35         3
## 4        1028       EDT   G     Control          C3           50         4
## 5        1019       EDT   G     Control          C6           10         5
## 6        1020       EDT   G     Control          C6           20         6
##   Dilution_Factor Field_.Notes CO2_Conc_ppm     CH4_Flag     CO2_Flag
## 1               1           NA     3199.083 Within Range Within Range
## 2               1           NA     5913.915 Within Range Within Range
## 3               1           NA    10094.383 Within Range Within Range
## 4               1           NA    15507.908 Within Range Within Range
## 5               1           NA     3668.013 Within Range Within Range
## 6               1           NA    13589.571 Within Range Within Range
##   CH4_Conc_ppm_dilcorr CO2_Conc_ppm_dilcorr CH4_Conc_umol CO2_Conc_umol
## 1             2.532985             3199.083     157.91680      199444.1
## 2             1.917852             5913.915     119.56684      368698.0
## 3             1.648335            10094.383     102.76403      629325.6
## 4             1.519918            15507.908      94.75798      966827.2
## 5             2.713720             3668.013     169.18457      228679.1
## 6             2.331640            13589.571     145.36411      847230.1
```

```r
dat$Depth <- dat$Sample_Depth
dat$Depth <- revalue(dat$Depth, c("10" = 10, "20" = 20, "35" = 35, "50"= 50, "ATM" = 0))
dat$Depth <- as.numeric(dat$Depth)
```

```r
soil_ch4 <- ggplot(data=dat, aes( x=CH4_Conc_ppm_dilcorr, y=Depth, color=Sample_Plot)) +
  geom_point(size=4)  + scale_y_reverse() + theme_bw()

soil_co2 <- ggplot(data=dat, aes( x=CO2_Conc_ppm_dilcorr, y=Depth, color=Sample_Plot)) +
  geom_point(size=4)  + scale_y_reverse() + theme_bw()

ggarrange(soil_ch4, soil_co2, nrow=1, ncol=2, common.legend = TRUE, legend="bottom")
```




```r
dat$Sample_Month <- as.factor(dat$Sample_Month)
dat$Sample_Depth <- as.factor(dat$Sample_Depth)
dat$Sample_Plot <- as.factor(dat$Sample_Plot)

levels(dat$Sample_Plot)
```

```
## [1] "Control"    "Freshwater" "Seawater"
```

```r
dat <- as.data.frame(dat)

ghg_avg <- dat %>%
  group_by(Sample_Depth, Sample_Plot) %>%
  dplyr::summarize(mean_CH4_ppm = mean(CH4_Conc_ppm_dilcorr, na.rm=TRUE),
                   sd_CH4 = sd(CH4_Conc_ppm_dilcorr, na.rm=TRUE),
                   mean_CO2_ppm = mean(CO2_Conc_ppm_dilcorr, na.rm=TRUE),
                   sd_CO2 = sd(CO2_Conc_ppm_dilcorr, na.rm=TRUE))
```

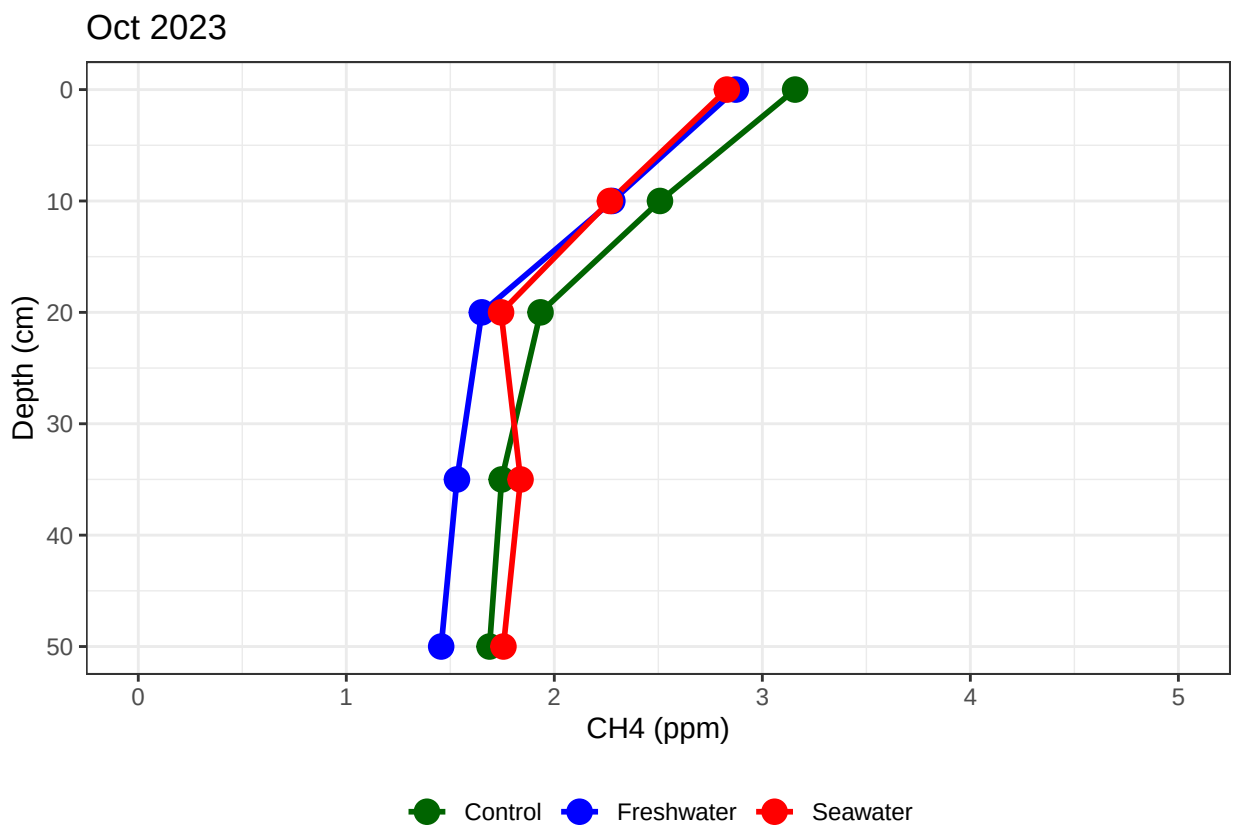
```
## `summarise()` has grouped output by 'Sample_Depth'. You can override using the
## `.groups` argument.
```

```r
head(ghg_avg)
```

```
## # A tibble: 6 x 6
## # Groups:   Sample_Depth [2]
##   Sample_Depth Sample_Plot mean_CH4_ppm sd_CH4 mean_CO2_ppm sd_CO2
##   <fct>        <fct>              <dbl>  <dbl>        <dbl>  <dbl>
## 1 10           Control             2.51 0.195         2983.   707.
## 2 10           Freshwater          2.28 0.136         2971.   421.
## 3 10           Seawater            2.27 0.312         5347.  4172.
## 4 20           Control             1.93 0.280         7922.  3247.
## 5 20           Freshwater          1.65 0.0978        7127.   984.
## 6 20           Seawater            1.74 0.395        13623.  3832.
```

```r
ghg_avg$Depth <- revalue(ghg_avg$Sample_Depth, c("10" = 10, "20" = 20, "35" = 35, "50"= 50, "ATM" = 0))
ghg_avg$Depth <- as.character(ghg_avg$Depth)
ghg_avg$Depth <- as.numeric(ghg_avg$Depth)

oct_avg_soil_ch4 <- ggplot(data=ghg_avg, aes(col=Sample_Plot, fill=Sample_Plot)) +
  geom_line(size=1, aes(Depth, mean_CH4_ppm)) +
  geom_point(size=4, aes(Depth, mean_CH4_ppm)) +
  scale_color_manual(values=c( "darkgreen","blue", "red")) +
  scale_fill_manual(values=c( "darkgreen","blue", "red")) +
  coord_flip() +
  scale_x_reverse() +
  ylim(0,5) +
  labs(title = "Oct 2023", y="CH4 (ppm)", x="Depth (cm)") +
  theme_bw() +
  theme(legend.position="bottom") +
  theme(legend.title=element_blank())
oct_avg_soil_ch4
```

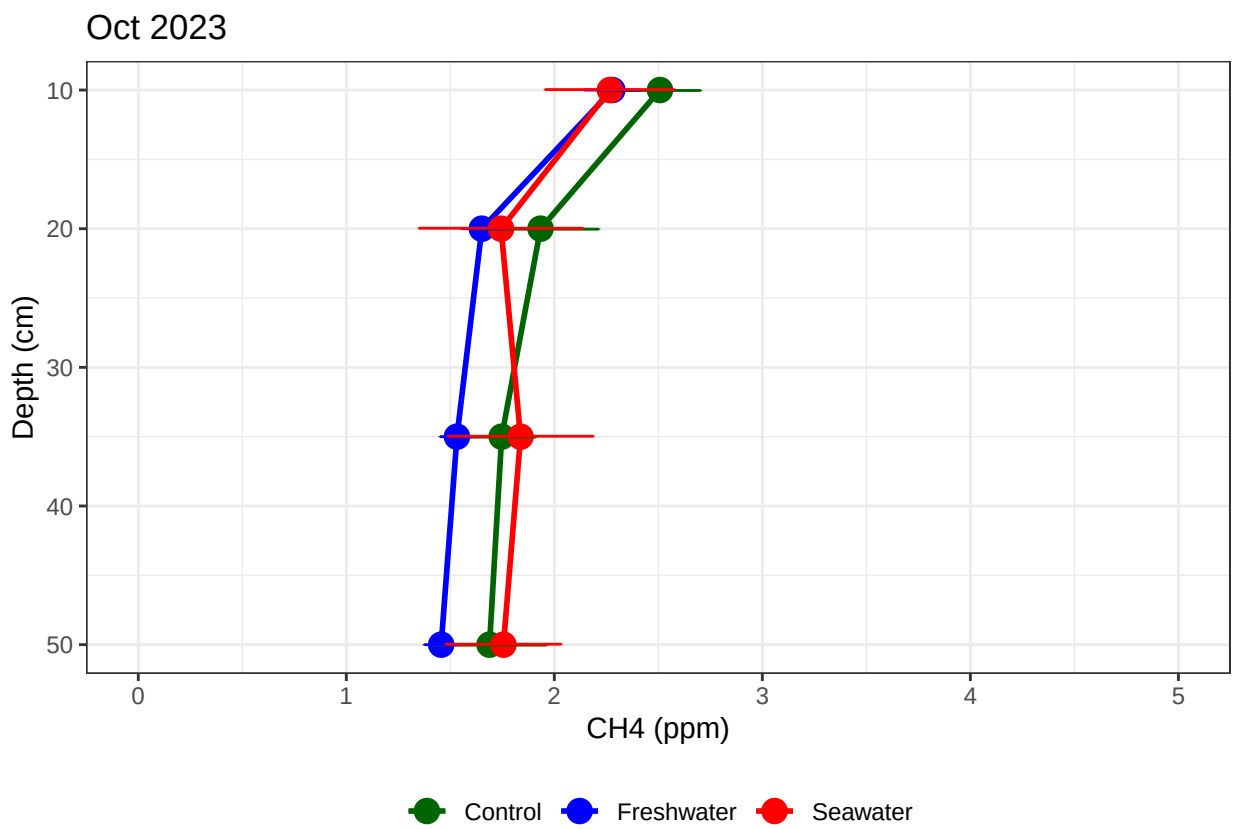



```
ghg_avg1 <-  ghg_avg %>%
  filter(!(Sample_Depth == "ATM"))
head(ghg_avg1)
```

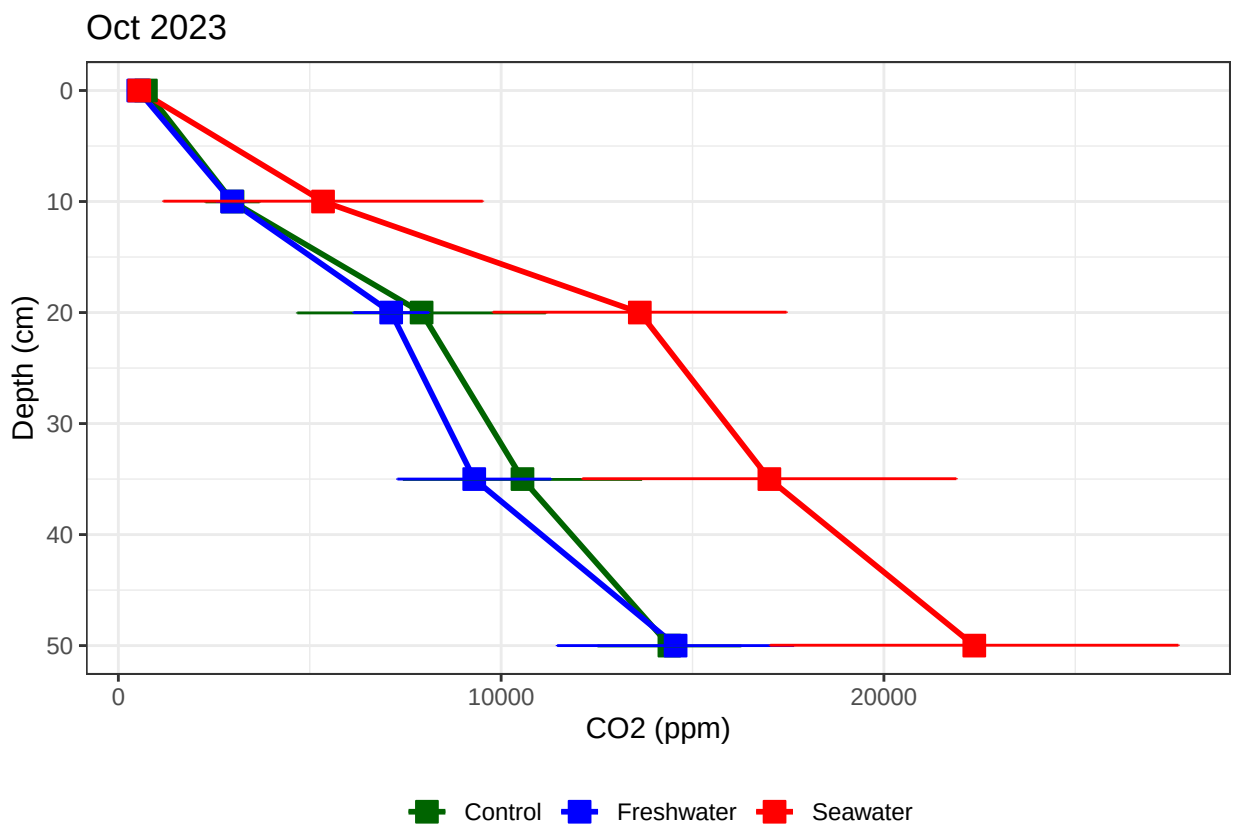
```
## # A tibble: 6 x 7
## # Groups:   Sample_Depth [2]
##   Sample_Depth Sample_Plot mean_CH4_ppm sd_CH4 mean_CO2_ppm sd_CO2 Depth
##   <fct>        <fct>              <dbl>  <dbl>        <dbl>  <dbl> <dbl>
## 1 10           Control             2.51 0.195         2983.   707.    10
## 2 10           Freshwater          2.28 0.136         2971.   421.    10
## 3 10           Seawater            2.27 0.312         5347.  4172.    10
## 4 20           Control             1.93 0.280         7922.  3247.    20
## 5 20           Freshwater          1.65 0.0978        7127.   984.    20
## 6 20           Seawater            1.74 0.395        13623.  3832.    20
```

```
oct_avg_soil_ch41 <- ggplot(data=ghg_avg1, aes(col=Sample_Plot, fill=Sample_Plot)) +
  geom_line(size=1, aes(Depth, mean_CH4_ppm)) +
  geom_point(size=4, aes(Depth, mean_CH4_ppm)) +
  scale_color_manual(values=c( "darkgreen","blue", "red")) +
  scale_fill_manual(values=c( "darkgreen","blue", "red")) +
  coord_flip() +
  scale_x_reverse() +
  ylim(0,5) +
  labs(title = "Oct 2023", y="CH4 (ppm)", x="Depth (cm)") +
  theme_bw() +
```

```
  theme(legend.position="bottom") +
  theme(legend.title=element_blank())+
  geom_errorbar(aes(Depth, ymin=mean_CH4_ppm-sd_CH4,
                    ymax=mean_CH4_ppm+sd_CH4),width=0.3,position=position_dodge(.1))
oct_avg_soil_ch41
```




```
oct_avg_soil_co21 <- ggplot(data=ghg_avg, aes(col=Sample_Plot, fill=Sample_Plot)) +
  geom_line(size=1, aes(Depth, mean_CO2_ppm)) +
  geom_point(shape=22, size=4, aes(Depth, mean_CO2_ppm)) +
  scale_color_manual(values=c( "darkgreen","blue", "red")) +
  scale_fill_manual(values=c( "darkgreen","blue", "red")) +
  coord_flip() +
  scale_x_reverse() +
  #ylim(0,5) +
  labs(title = "Oct 2023", y="CO2 (ppm)", x="Depth (cm)") +
  theme_bw() +
  theme(legend.position="bottom") +
  theme(legend.title=element_blank())+
  geom_errorbar(aes(Depth, ymin=mean_CO2_ppm-sd_CO2,
                    ymax=mean_CO2_ppm+sd_CO2),width=0.3,position=position_dodge(.1))
oct_avg_soil_co21
```




```
##########################################################################################
```

November 2023

```
#### NOV 2023 ####
##########################################################################################
#read in some data
dat <- read.csv("2023 Processed Data/COMPASS_TEMPEST_SGW_202311_Processed.csv")
head(dat)
```

```
##   X   Machine       User Run_Date Sample_Year Sample_Month Sample_Day
## 1 1 Varian GC Max Wegner 20240109        2023     November          3
## 2 2 Varian GC Max Wegner 20240109        2023     November          3
## 3 3 Varian GC Max Wegner 20240109        2023     November          3
## 4 4 Varian GC Max Wegner 20240109        2023     November          3
## 5 5 Varian GC Max Wegner 20240109        2023     November          3
## 6 6 Varian GC Max Wegner 20240109        2023     November          3
##   Sample_Time Time_Zone G_W Sample_Plot Sample_Grid Sample_Depth Sample_ID
## 1        1403       EDT   G     Control          C3           10         1
## 2        1404       EDT   G     Control          C3           20         2
## 3        1405       EDT   G     Control          C3           35         3
## 4        1406       EDT   G     Control          C3           50         4
## 5        1412       EDT   G     Control          C6           10         5
```

```
## 6        1413       EDT   G     Control          C6           20         6
##   Dilution_Factor Field_Notes CO2_Conc_ppm     CH4_Flag     CO2_Flag
## 1               1          NA     1560.961 Within Range Within Range
## 2               1          NA     3106.438 Within Range Within Range
## 3               1          NA     4560.757 Within Range Within Range
## 4               1          NA     8915.923 Within Range Within Range
## 5               1          NA     2083.328 Within Range Within Range
## 6               1          NA     4953.173 Within Range Within Range
##   CH4_Conc_ppm_dilcorr CO2_Conc_ppm_dilcorr CH4_Conc_umol CO2_Conc_umol
## 1             2.223527             1560.961     138.62386      97316.75
## 2             1.626635             3106.438     101.41116     193668.18
## 3             1.558555             4560.757      97.16674     284336.46
## 4             1.471475             8915.923      91.73783     555855.53
## 5             2.391353             2083.328     149.08685     129883.29
## 6             1.959121             4953.173     122.13972     308801.33
```

```
dat <-  dat %>%
  filter(!str_detect(Sample_Plot, "Transition"))
head(dat)
```

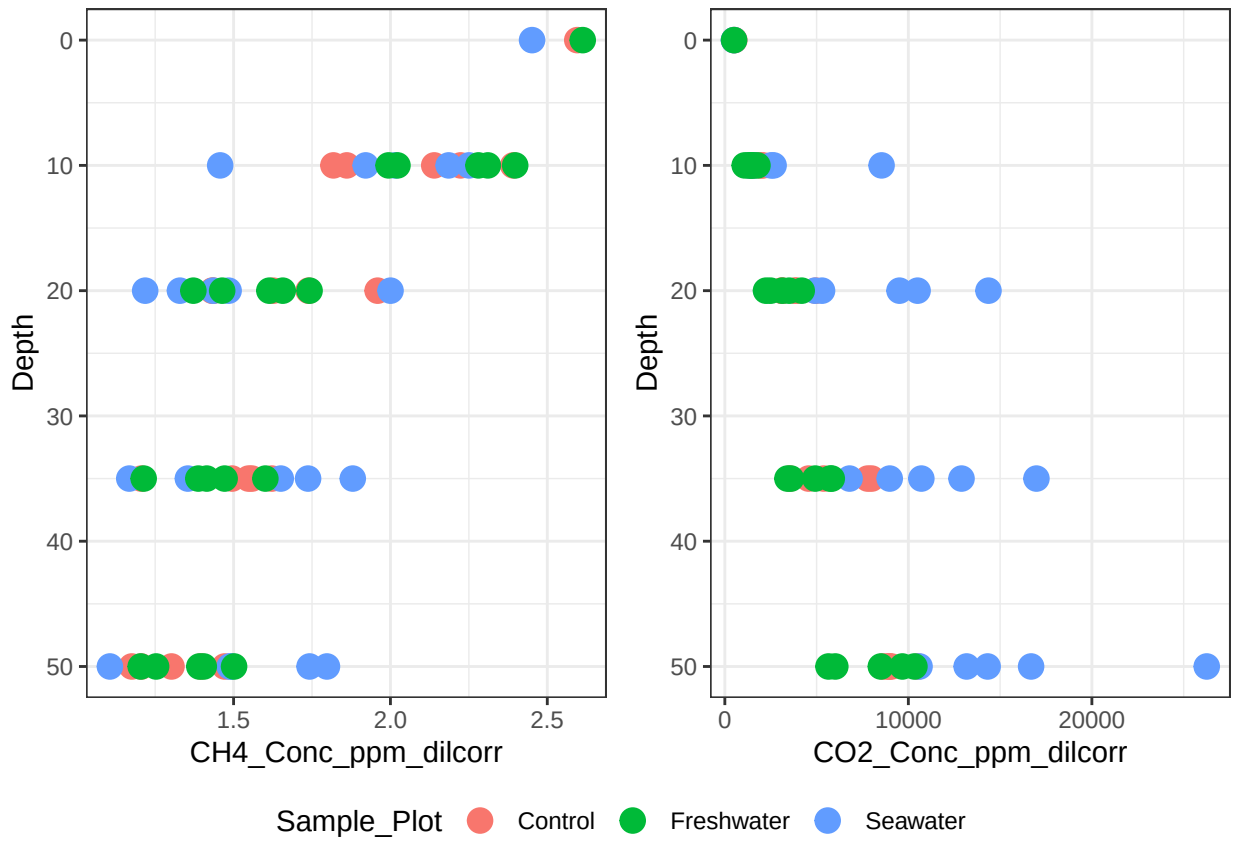
```
##   X   Machine       User Run_Date Sample_Year Sample_Month Sample_Day
## 1 1 Varian GC Max Wegner 20240109        2023     November          3
## 2 2 Varian GC Max Wegner 20240109        2023     November          3
## 3 3 Varian GC Max Wegner 20240109        2023     November          3
## 4 4 Varian GC Max Wegner 20240109        2023     November          3
## 5 5 Varian GC Max Wegner 20240109        2023     November          3
## 6 6 Varian GC Max Wegner 20240109        2023     November          3
##   Sample_Time Time_Zone G_W Sample_Plot Sample_Grid Sample_Depth Sample_ID
## 1        1403       EDT   G     Control          C3           10         1
## 2        1404       EDT   G     Control          C3           20         2
## 3        1405       EDT   G     Control          C3           35         3
## 4        1406       EDT   G     Control          C3           50         4
## 5        1412       EDT   G     Control          C6           10         5
## 6        1413       EDT   G     Control          C6           20         6
##   Dilution_Factor Field_Notes CO2_Conc_ppm     CH4_Flag     CO2_Flag
## 1               1          NA     1560.961 Within Range Within Range
## 2               1          NA     3106.438 Within Range Within Range
## 3               1          NA     4560.757 Within Range Within Range
## 4               1          NA     8915.923 Within Range Within Range
## 5               1          NA     2083.328 Within Range Within Range
## 6               1          NA     4953.173 Within Range Within Range
##   CH4_Conc_ppm_dilcorr CO2_Conc_ppm_dilcorr CH4_Conc_umol CO2_Conc_umol
## 1             2.223527             1560.961     138.62386      97316.75
## 2             1.626635             3106.438     101.41116     193668.18
## 3             1.558555             4560.757      97.16674     284336.46
## 4             1.471475             8915.923      91.73783     555855.53
## 5             2.391353             2083.328     149.08685     129883.29
## 6             1.959121             4953.173     122.13972     308801.33
```

```
dat$Depth <- dat$Sample_Depth
dat$Depth <- revalue(dat$Depth, c("10" = 10, "20" = 20, "35" = 35, "50"= 50, "ATM" = 0))
dat$Depth <- as.numeric(dat$Depth)
```

```r
soil_ch4 <- ggplot(data=dat, aes( x=CH4_Conc_ppm_dilcorr, y=Depth, color=Sample_Plot)) +
  geom_point(size=4)  + scale_y_reverse() + theme_bw()

soil_co2 <- ggplot(data=dat, aes( x=CO2_Conc_ppm_dilcorr, y=Depth, color=Sample_Plot)) +
  geom_point(size=4)  + scale_y_reverse() + theme_bw()

ggarrange(soil_ch4, soil_co2, nrow=1, ncol=2, common.legend = TRUE, legend="bottom")
```




```r
dat$Sample_Month <- as.factor(dat$Sample_Month)
dat$Sample_Depth <- as.factor(dat$Sample_Depth)
dat$Sample_Plot <- as.factor(dat$Sample_Plot)

levels(dat$Sample_Plot)
```

```
## [1] "Control"    "Freshwater" "Seawater"
```

```r
dat <- as.data.frame(dat)

ghg_avg <- dat %>%
  group_by(Sample_Depth, Sample_Plot) %>%
  dplyr::summarize(mean_CH4_ppm = mean(CH4_Conc_ppm_dilcorr, na.rm=TRUE),
                   sd_CH4 = sd(CH4_Conc_ppm_dilcorr, na.rm=TRUE),
                   mean_CO2_ppm = mean(CO2_Conc_ppm_dilcorr, na.rm=TRUE),
                   sd_CO2 = sd(CO2_Conc_ppm_dilcorr, na.rm=TRUE))
```

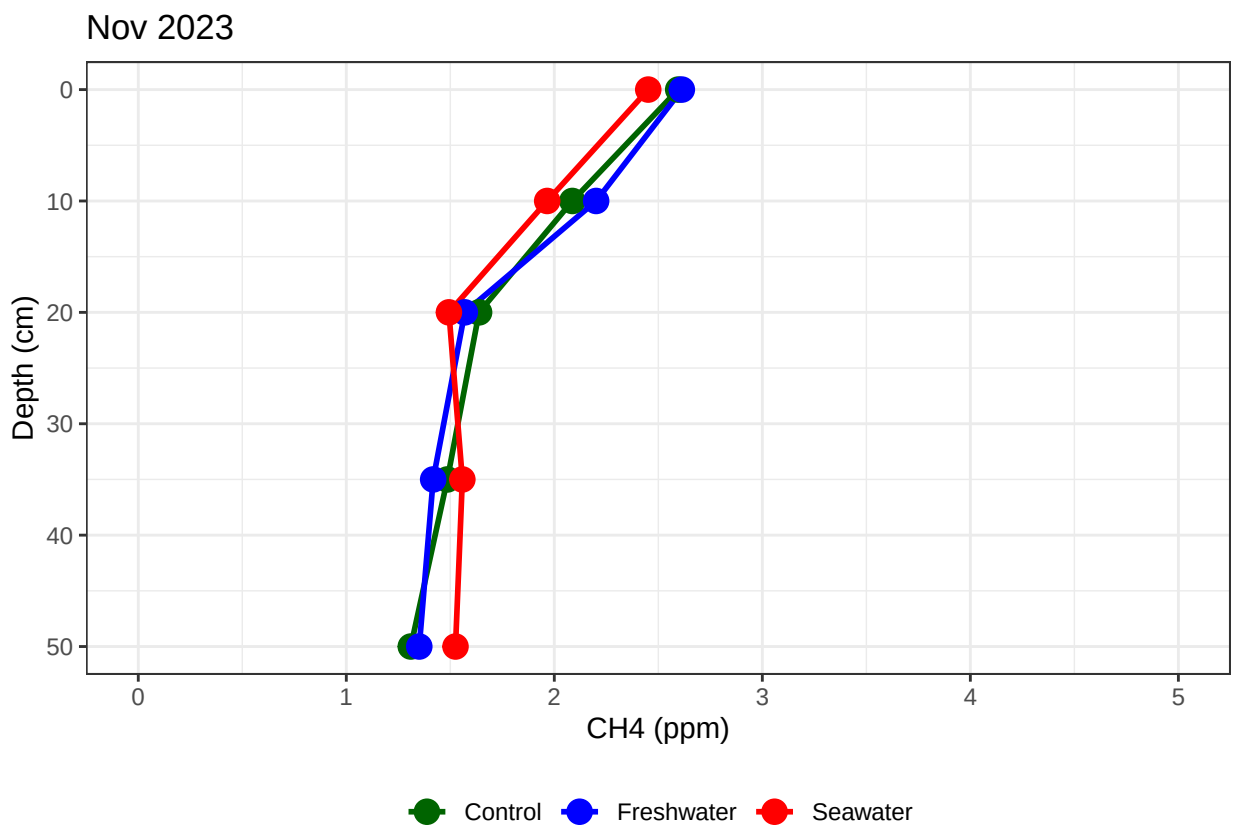
```
## `summarise()` has grouped output by 'Sample_Depth'. You can override using the
## `.groups` argument.
```

```r
head(ghg_avg)
```

```
## # A tibble: 6 x 6
## # Groups:   Sample_Depth [2]
##   Sample_Depth Sample_Plot mean_CH4_ppm sd_CH4 mean_CO2_ppm sd_CO2
##   <fct>        <fct>              <dbl>  <dbl>        <dbl>  <dbl>
## 1 10           Control             2.09  0.244        1679.   312.
## 2 10           Freshwater          2.20  0.182        1432.   256.
## 3 10           Seawater            1.97  0.313        3324.  2973.
## 4 20           Control             1.64  0.222        3776.   755.
## 5 20           Freshwater          1.57  0.150        3114.   784.
## 6 20           Seawater            1.49  0.301        8920.  3927.
```

```r
ghg_avg$Depth <- revalue(ghg_avg$Sample_Depth, c("10" = 10, "20" = 20, "35" = 35, "50"= 50, "ATM" = 0))
ghg_avg$Depth <- as.character(ghg_avg$Depth)
ghg_avg$Depth <- as.numeric(ghg_avg$Depth)

nov_avg_soil_ch4 <- ggplot(data=ghg_avg, aes(col=Sample_Plot, fill=Sample_Plot)) +
  geom_line(size=1, aes(Depth, mean_CH4_ppm)) +
  geom_point(size=4, aes(Depth, mean_CH4_ppm)) +
  scale_color_manual(values=c( "darkgreen","blue", "red")) +
  scale_fill_manual(values=c( "darkgreen","blue", "red")) +
  coord_flip() +
  scale_x_reverse() +
  ylim(0,5) +
  labs(title = "Nov 2023", y="CH4 (ppm)", x="Depth (cm)") +
  theme_bw() +
  theme(legend.position="bottom") +
  theme(legend.title=element_blank())
nov_avg_soil_ch4
```

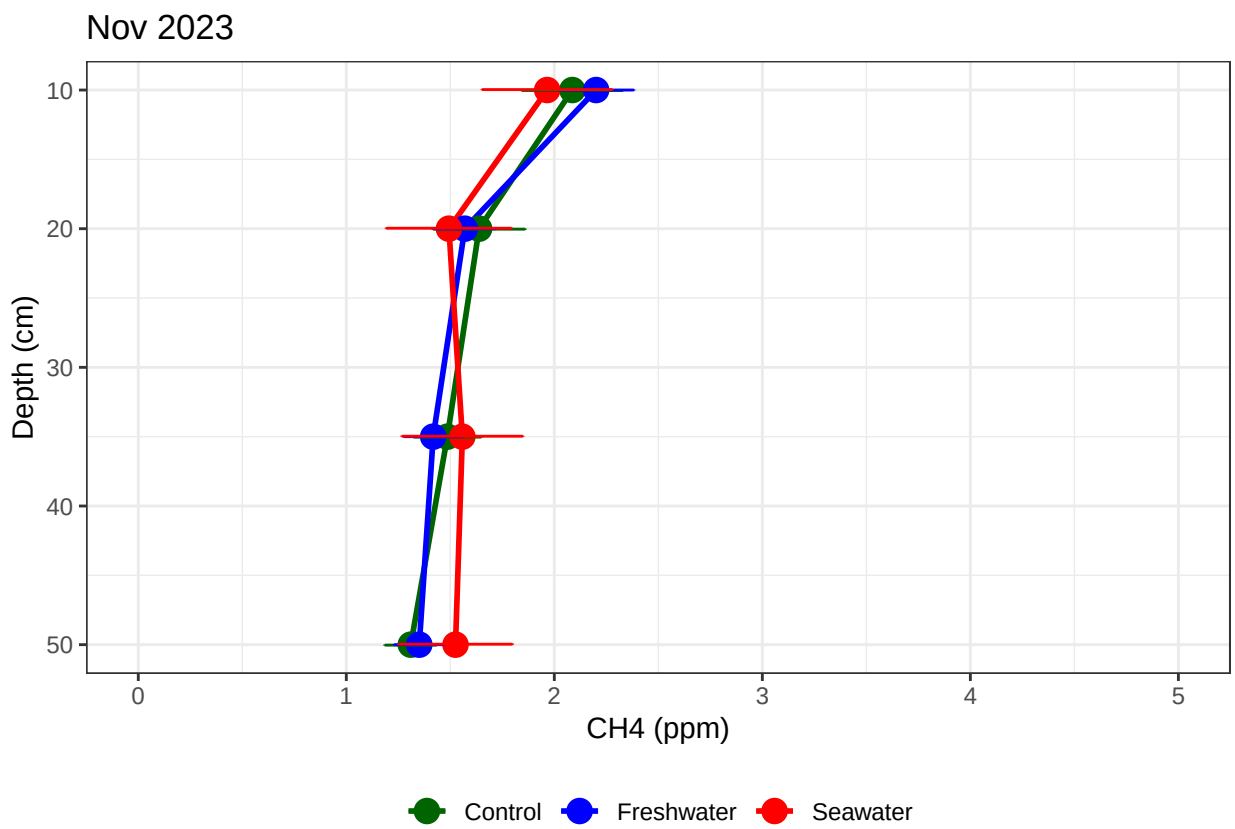



```
ghg_avg1 <-  ghg_avg %>%
  filter(!(Sample_Depth == "ATM"))
head(ghg_avg1)
```

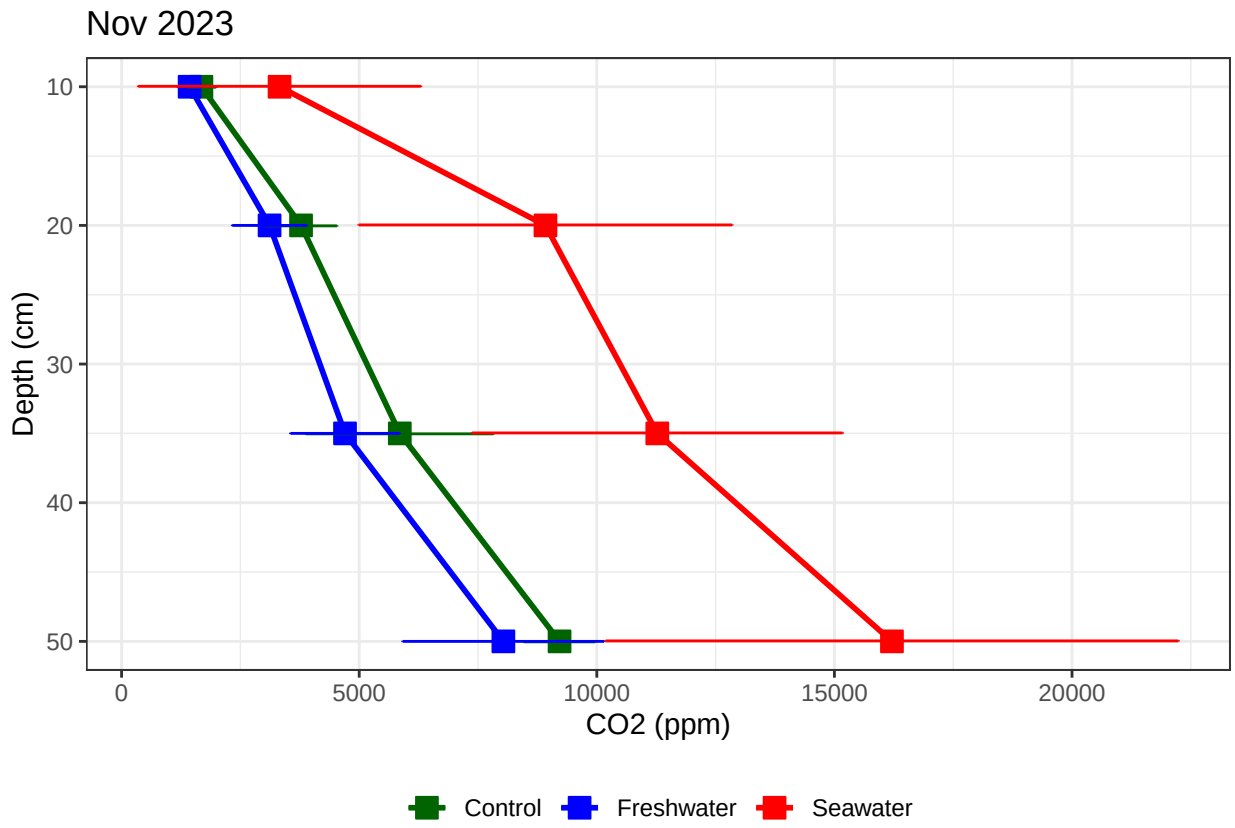
```
## # A tibble: 6 x 7
## # Groups:   Sample_Depth [2]
##   Sample_Depth Sample_Plot mean_CH4_ppm sd_CH4 mean_CO2_ppm sd_CO2 Depth
##   <fct>        <fct>              <dbl>  <dbl>        <dbl>  <dbl> <dbl>
## 1 10           Control             2.09  0.244        1679.   312.    10
## 2 10           Freshwater          2.20  0.182        1432.   256.    10
## 3 10           Seawater            1.97  0.313        3324.  2973.    10
## 4 20           Control             1.64  0.222        3776.   755.    20
## 5 20           Freshwater          1.57  0.150        3114.   784.    20
## 6 20           Seawater            1.49  0.301        8920.  3927.    20
```

```
nov_avg_soil_ch41 <- ggplot(data=ghg_avg1, aes(col=Sample_Plot, fill=Sample_Plot)) +
  geom_line(size=1, aes(Depth, mean_CH4_ppm)) +
  geom_point(size=4, aes(Depth, mean_CH4_ppm)) +
  scale_color_manual(values=c( "darkgreen","blue", "red")) +
  scale_fill_manual(values=c( "darkgreen","blue", "red")) +
  coord_flip() +
  scale_x_reverse() +
  ylim(0,5) +
  labs(title = "Nov 2023", y="CH4 (ppm)", x="Depth (cm)") +
  theme_bw() +
```

```
  theme(legend.position="bottom") +
  theme(legend.title=element_blank())+
  geom_errorbar(aes(Depth, ymin=mean_CH4_ppm-sd_CH4,
                    ymax=mean_CH4_ppm+sd_CH4),width=0.3,position=position_dodge(.1))
nov_avg_soil_ch41
```




```
nov_avg_soil_co21 <- ggplot(data=ghg_avg1, aes(col=Sample_Plot, fill=Sample_Plot)) +
  geom_line(size=1, aes(Depth, mean_CO2_ppm)) +
  geom_point(shape=22, size=4, aes(Depth, mean_CO2_ppm)) +
  scale_color_manual(values=c( "darkgreen","blue", "red")) +
  scale_fill_manual(values=c( "darkgreen","blue", "red")) +
  coord_flip() +
  scale_x_reverse() +
  #ylim(0,5) +
  labs(title = "Nov 2023", y="CO2 (ppm)", x="Depth (cm)") +
  theme_bw() +
  theme(legend.position="bottom") +
  theme(legend.title=element_blank())+
  geom_errorbar(aes(Depth, ymin=mean_CO2_ppm-sd_CO2,
                    ymax=mean_CO2_ppm+sd_CO2),width=0.3,position=position_dodge(.1))
nov_avg_soil_co21
```




```
##########################################################################################
```

Viz Together?